

SOUTH



AUSTRALIA.

DEPARTMENT OF MINES.

Geological Survey of South
Australia.

BULLETIN No. 9.

*The Iron Ore Resources of South
Australia.*

By R. LOCKHART JACK, B.E., F.G.S., Deputy Government Geologist.

Issued under the authority of

The Honorable T. PASCOE, M.L.C., Minister of Mines.

ADELAIDE:

R. E. E. ROGERS, GOVERNMENT PRINTER, NORTH TERRACE.

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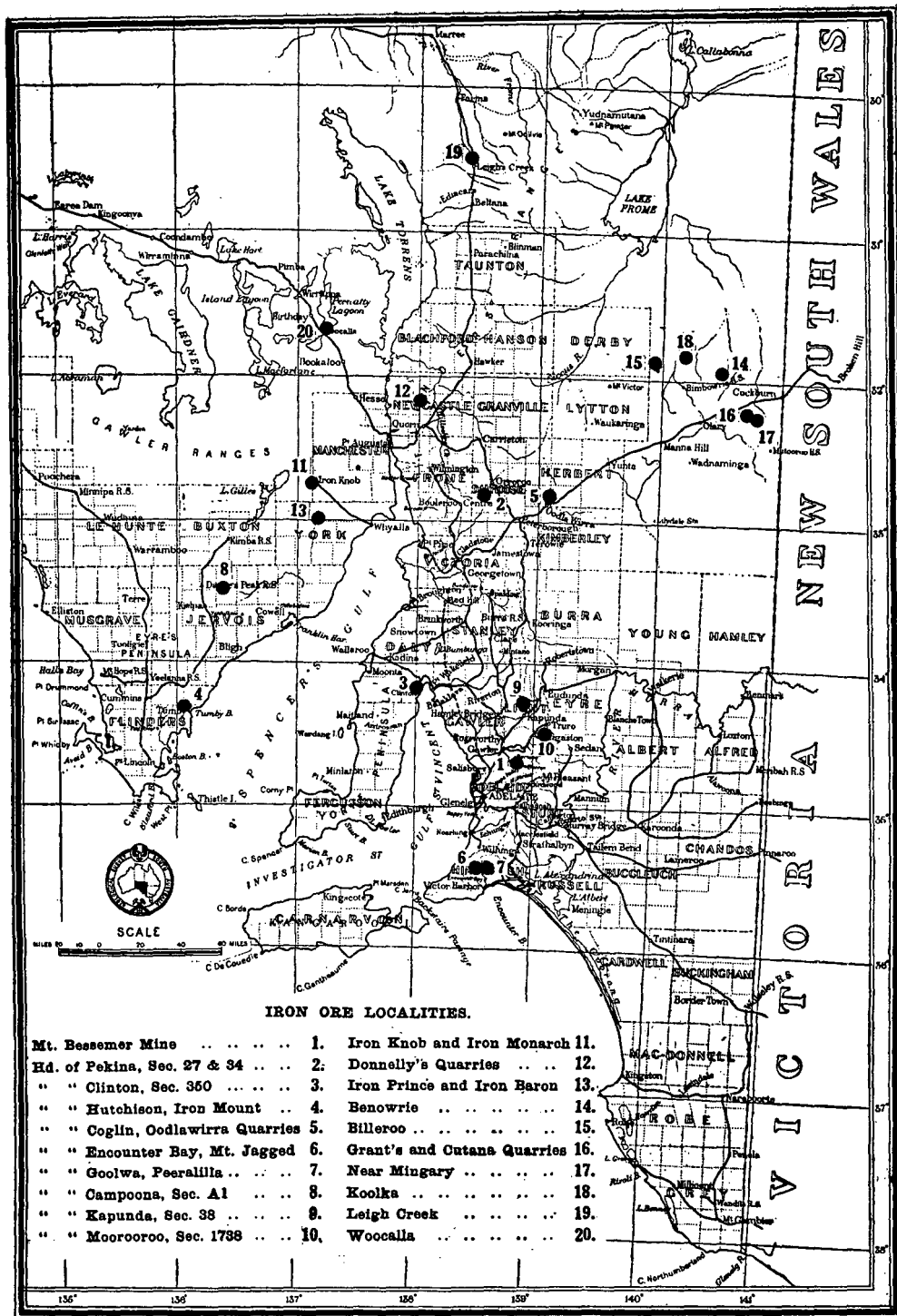
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Portion of South Australia showing Localities of Iron Deposits described.

CONTENTS.

	PAGE.
LETTERS OF TRANSMITTAL	5 and 6
AUTHORITY	5
INTRODUCTION	7
UTILIZATION	7
VALUE OF IRON ORE	9
GENERAL	10
PRODUCTION	11
ECONOMIC IRON MINERALS	12
CONCLUSION	13
DETAILS OF DEPOSITS—	
County Adelaide—	
Hundred of Barossa, Mount Bessemer Mine	15
County of Dalhousie—	
Hundred of Pekina, section 27	20
Hundred of Pekina, section 34	20
County of Daly	
Hundred of Clinton, section 350	21
County of Flinders—	
Hundred of Hutchison, Iron Mount Mine	21
County of Herbert—	
Hundred of Coglein, sections 83, 127, and 128, Oodlawirra flux	22
County of Hindmarsh—	
Hundred of Encounter Bay, section 52, Mount Jagged	24
Hundred of Goolwa, section 264, Peeralilla	26
County of Jervois—	
Hundred of Campoona, section A1	28
County of Light—	
Hundred of Kapunda, section 38	28
Hundred of Moorooroo, section 1738	29
County of Manchester—	
Iron Knob and Iron Monarch	30
Holding	30
Locality	30
Geology	30
Ore bodies	33
Depth of ore body and tonnage	36
Ore types	38
Waste zones	39
Grade	40
Methods	42
Conclusion	42

DETAILS OF DEPOSITS— <i>continued.</i>		PAGE.
County of Newcastle—		
Hundred of Yarrah, section F, Donnelly's or Comstock Quarries		44
County of York—		
Middleback Range, Iron Prince and Iron Baron leases		48
Beyond Counties—		
Benowrie or Boolcoomatta Station		58
Billeroo Ironstone Deposits		60
Grant's and Cutana Quarries		61
Mingary, Mineral Surveys 19388 and 19389, near		66
Koolka		67
Leigh Creek Coal Measures		69
Woocalla, Iron Ore		70
Other localities (list)		71

PLATES		
Locality Plan		Frontispiece
Iron Knob and Iron Monarch		} At end
Iron Monarch		
Iron Prince and Iron Baron		

LETTER OF TRANSMITTAL.

Geological Survey of South Australia,

Adelaide, February 6th, 1922.

Sir—I have the honor to transmit to you the report of Mr. R. Lockhart Jack, B.E., F.G.S., Deputy Government Geologist, on the Iron Ore Resources of South Australia.

This report has been prepared as one of the series of departmental publications dealing with the mineral resources of South Australia. Already there have been published bulletins dealing with the mining district of Wallaroo and Moonta, the phosphate deposits, and the salt and gypsum resources of the State.

Iron ore has long been produced from South Australian deposits for use as a flux in smelting operations, and the rate of production has been increased enormously with the establishment of modern steel works at Newcastle, New South Wales. Hence it has come to pass that the value of the iron ore produced in South Australia exceeded, in 1919, the value of any other mineral product. This position of supremacy is likely to be maintained in the future, if economic conditions permit the successful operation of the mine, the smelting works, and the secondary industries which depend on supplies of pig iron and steel.

I have, &c.,

L. KEITH WARD, Government Geologist.

The Honorable T. Pascoe, M.L.C., Minister of Mines.

Submitted for approval to print as a Bulletin of the Geological Survey of South Australia.

Approved,

T. PASCOE, Minister of Mines.

February 7th, 1922.

To the Government Geologist.

Sir—I enclose herewith my report on the Iron Ore Resources of South Australia.

All the larger known bodies have been examined and described, and many reputed to be of some magnitude have been found to be insignificant in size.

Other deposits, of which a list is given, are known to exist, but, as far as could be ascertained, are of little size or distant from railways, and so were not inspected.

The outstanding deposits are the Iron Knob and Iron Monarch, and the Iron Prince and Iron Baron, in the Middleback Range, estimated to have ore reserves above the level of the surrounding country of 133,000,000 and 32,000,000 tons. Portions of the Iron Monarch deposit are highly manganeseiferous, and, owing to the demands of the furnaces for ore yielding less than 1 per cent. of manganese in the pig iron, this manganese is a disability which has to be met by blending with the non-manganeseiferous portions of the ore bodies. For this reason the ores of Middleback will be of especial value to the Broken Hill Proprietary Company, which controls both these great deposits on which is based the great industrial undertaking at Newcastle.

I wish to record my indebtedness to the staff of the Broken Hill Proprietary Company for facilities and information given me during my examination of the deposits controlled by them.

I have the honor to be, Sir,

Your obedient servant,

R. LOCKHART JACK, Deputy Government Geologist.

Adelaide, February 6th, 1922.

IRON ORE RESOURCES OF SOUTH AUSTRALIA.

In any discussion of the iron ore resources of South Australia it has always to be borne in mind that, in a country of great distances, with in many cases complete absence of transport facilities, bodies of iron ore are valueless when the cost of transport becomes important. Consequently, outlying or inaccessible deposits are not always recorded by the discoverer, and as settlement and transport facilities increase, so will the number of localities in which iron deposits are known to be augmented.

The discovery of the mines of Broken Hill and the erection of lead smelters at Broken Hill, Port Pirie, Dry Creek, and Port Adelaide, caused a demand for ironstone flux which was met by the exploitation of many deposits throughout the country. Size was of less consequence than reasonable purity and ease of transport, as the raiser found a ready market for even limited quantities. The effect of the closing down of the scattered smelters and the concentration of effort at Port Pirie led to the abandonment of many of the flux workings, and when the Iron Knob was connected by a tramway to the port of Hummocky Hill, now known as Whyalla, in 1903, the ease of supplying very high-grade hæmatite and its low cost caused the cessation of work in all the other ironstone quarries. How widespread was the call for flux, whether for lead or copper smelting, may be gathered from the following list in which all iron ore localities known to the Department of Mines are recorded. Those that are known to have supplied flux are italicised, while those that are described later are shown by an asterisk.

*Angaston**, *Barossa**, Beltana, Billeroo*, Burra, Birdwood, Clinton*, Cutana*, *Crystal Brook*, Eureka, Gladstone, Gum Creek, Huddleston, Hawker (E. of), Hundred of Hawker, *Iron Knob**, *Iron Monarch**, Koolka*, Kapunda, Kersbrook, Mount Jagged*, Normanville, Oodlawirra*, Peerabilla*, Pernatty Lagoon*, Pekina*, Pine Point, Port Lincoln, Pandurra, Para Wirra, Quorn (Donnelly's)*, Talunga.

While the majority of these deposits yielded or could yield flux, many are, by reason of their size, grade, and presence of deleterious constituents, unsuited for the manufacture of iron. Others again are ruled out by reason of their inaccessibility, a term which must be taken to include the expense of bringing the ore to fuel—that is to the seaboard, or fuel from the seaboard to the ore.

USES.

The uses to which deposits may be turned are:—(a) Steel making, (b) foundry pig, (c) flux, (d) ochres, and (e) road material; and the requirements for each class are briefly set out below.

(a) *Steel Making*.—The high cost of blast furnace plant and mill and the great capacity of a modern blast furnace are such that it is considered that a working-life of not less than 25 years must be counted upon as a minimum before any deposit can receive serious consideration for this purpose. Modern blast furnaces produce as much as 180,000 tons of pig annually,

and the average output of pig iron per furnace in 1916, based on the total production of pig iron and the number (448) of blast furnaces in the United States*, was 88,000 tons.

This average rate of production would call for about 150,000 tons of 60 per cent. ore annually, equivalent to a deposit of 3,750,000 tons, on a 25 years' life for the plant. Examination of the records of a large number of plants shows that they control much larger reserves than are indicated by this calculation. It may be noted that the Broken Hill Proprietary Company's steel works, operating three blast furnaces, are capable of producing about 450,000 tons of pig annually, and that the cost of the whole plant, which, however, is designed with a view to the erection of further blast furnaces, was £5,039,107 (exclusive of the depreciation written off) on May 31st, 1921.

From these figures it may be seen that the ore body justifying the erection of a modern steel-making plant must be a large one.

In addition to tonnage, other conditions must be favorable before a deposit can be considered suitable for steel work, viz.:—Relation to fuel, flux, and market for the product, and whether the site for the proposed works is such that the cost of assembling coal, flux, and ore, and of placing the product on the market, is the minimum possible. The Broken Hill Proprietary, for example, working the largest known iron ore body in South Australia—and probably in Australia—decided that Newcastle, New South Wales, best fulfilled these conditions, and established their works there, near the centres of coal production and greatest metal consumption.

Of very great commercial importance is the nature of the impurities in the iron ore. Silica, alumina, lime, magnesia, and others of less importance can be removed in the slag at additional cost for flux and fuel, but do not affect the quality of iron produced. On the other hand, certain elements, though present in small amount, enter the iron almost wholly, and affect it very definitely. Phosphorus, sulphur, titanium, and manganese may be cited, although under certain conditions manganiferous pig is intentionally produced. Titanium is regarded as undesirable in excess of 1 per cent. in the ore, as its elimination in the furnace requires more fuel and reduces the rate of smelting. Phosphorus is regarded as very deleterious, unless present in large amount, when it can be turned to account by the manufacture of basic slag fertiliser as a by-product. The percentage present in the pig iron determines the process that has to be adapted in the manufacture of steel.

The following table from Eckel (Iron Ores, p. 151) shows the allowable phosphorus in pig iron for various utilisations:—

Acid open hearth—Less than .05 per cent.

Acid Bessemer—Less than .1 per cent.

Basic open hearth for normal process—Not over 1.5 per cent. and, preferably, not over 1 per cent.; for special processes, 1.5 per cent. and over.

Basic Bessemer—At least 1.5 per cent., preferably over 2 per cent.

Foundry Iron—Wide range, according to special use of iron.

From this summary it may be seen that the requirements for steel making are strict, and include size of deposit, relation of deposit to fuel and market, and chemical composition of ore.

(b) *Foundry Pig*.—On the other hand, if it is intended to make and market pig iron only, the capital cost is much reduced, the amount to be set

* Mineral Industry, 1916, p. 403.

aside for amortisation of plant is less, and consequently a smaller deposit can be utilised. There is also less expensive supervision required, and overhead charges are reduced, consequently a smaller blast furnace than would be required for a steel plant can be used. This would reduce output and redemption charges, but would increase the cost of production per ton. The balancing of the two latter items, together with the size of the ore body available, would determine the size of the plant for the production of pig iron, to be sold as such. The presence of phosphorus would be less deleterious than in the case of steel making, as it increases fluidity for casting, though at the expense of strength. Here again the location of the works is a matter of calculation to ensure that the cost of assembling of the raw materials and the distribution of the product is reduced to a minimum.

(c) *Flux*.—As already stated, the demand for flux for the reduction of lead or copper ores resulted in the opening of very numerous quarries throughout the State. The essentials were not onerous, comparative freedom from silica and an iron content of not less than 50 per cent. being all that were called for. Transport was the chief limiting factor, and size of deposit was of little importance, as, provided a few truck loads could be mined economically from any ore body, it could be sold. The development of Iron Knob has resulted in a large supply of even-grade flux, and at the present time, and indeed for years past, there has not been a single producer of flux other than the Iron Knob.

(d) *Ochres*.—Certain of the limonitic iron deposits are being utilised for ochres, both for the production of paint and for the purification of town gas. Soft material has hitherto been used, the raw yellow and various shades of brown and natural reds being mined. Some of the yellows are being calcined for the production of red. Ochres vary considerably in iron content, but in general it may be said that the staining power is proportional to the percentage of iron oxide present.

Owing to the comparatively high value of ochre and the relatively small tonnage required, it is less affected by locality and transport costs, and bodies of comparatively small size can be utilised.

(e) *Road Material*.—Where deposits of limonitic iron ores are available it has been found that the limonite produces an excellent road surface, owing to the property of cementation which it possesses.

In certain districts, notably the wetter pre-Cambrian areas around Port Lincoln and in the Adelaide hills, superficial deposits of nodular and pisolitic limonite of lateritic habit occur over considerable areas to a depth of 2ft. to 3ft. or even more, and locally are largely utilised for road making. The material is too soft for very heavy and continuous traffic, but might well be utilised as a cementing blinding on quartzite roadbeds wherever it can be obtained at reasonable cost.

On section 38, Kapunda, a fairly extensive vein of limonite has been quarried to furnish road metal for the adjoining roads with very satisfactory results. The use of limonite may also be observed in many parts of the country.

VALUE OF IRON ORE.

A good deal of misconception appears to be present locally as to the monetary value of ores of iron. Iron ore prices are ultimately based upon either demand or cost of production. As the iron smelting industry of Australia is at present constituted, there are two works in New South Wales and one projected in Queensland. Of the New South Wales plants, one at

Lithgow, in the Blue Mountains, depends on ores raised in New South Wales; the other at Newcastle draws its supplies from the Iron Knob, in South Australia.

There is a blast furnace at Mittagong, in New South Wales; but it has not been operated for many years, and only produced pig iron.

The site of the Queensland plant has been announced as at Bowen, and press reports have stated that ore will be drawn from Mount Biggenden (near Maryborough) and Mount Leviathan (at Cloncurry). The latter deposit is large, and of good quality, but is 481 miles by rail from seaboard. The Queensland Government has also secured control of very large and high-grade ironstone deposits at Yampi Sound, on the north-west coast of Western Australia, from which it is intended to draw supplies for the smelters.

It is thus obvious that the demand for South Australian ore, other than that of Iron Knob, is not likely to be serious, unless the cost f.o.b. approximates the cost of mining and placing on ship of Iron Knob ore or of the ores used in other plants. So long as the Newcastle Works have their own ample deposits to draw upon they cannot be expected to give a price above that at which they can raise ore themselves, and this price is below what may be called the commercial valuation of the ore. Such commercial valuation only obtains when there is competition between numerous smelters for a reasonable proportion of the ore that they require. This is exemplified in the case of two markets, called by Eckel the "Lake Superior Market," which for 1903-1913 averaged 7 cents ($3\frac{1}{4}$ d.) per unit of iron, and the "Atlantic Market." In the "Atlantic Market," which includes all ore sold at the consuming ports of the North Atlantic, prices ranged from $5\frac{1}{2}$ cents ($2\frac{1}{4}$ d.) to $8\frac{1}{2}$ cents ($4\frac{1}{4}$ d.) per unit of iron, depending on quality. These rates show an average value for 60 per cent. ore of approximately 20s. per ton, but the price in the last few years has been considerably higher. It is thus apparent that, even assuming a "buying" market to exist, the price of, say, 60 per cent. South Australian iron ores cannot be such that, under modern economic conditions, an export trade beyond Australia can be established.

GENERAL.

From consideration of the details of deposits shown in Part II. it may be seen that the deposits that afford a reasonable chance of yielding three to four million tons of ore within economic mining limits are very few, and, further, that their geographical relation to the railway system of the State and to the seaboard reduces the number that can reasonably be expected to maintain a steel works to two.

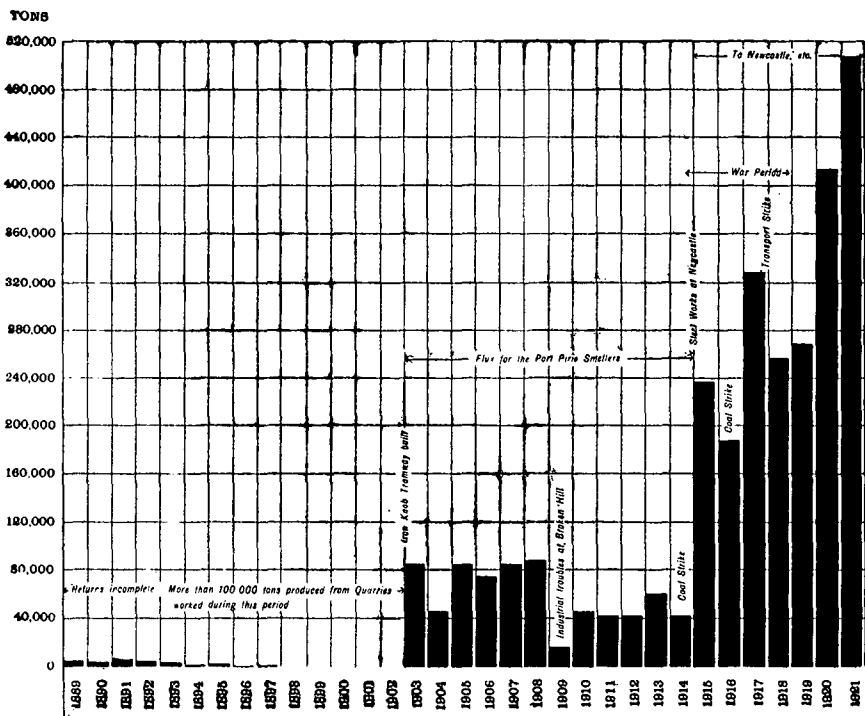
Combining a number of the small deposits with a central South Australian plant will not do much to solve the problem, as it seems probable that, until the steel requirements of South Australia are such as to absorb the bulk of steel produced, it will be sounder, financially, to take the ore to the coal than the reverse.

For the production of foundry pig, smaller and less pure deposits would suffice, and it is possible that, with the local consumption of the product, it would be economically sound to bring the coal to the ore. In point of magnitude Mount Bessemer, Donnelly's, Cutana, Billeroo, and Koolka might fall within the ambit of pig production. Some of these, however, are eliminated by cost of carriage.

The high phosphorus content of some of the limonitic bodies immediately suggests the basic process for the production of basic slag and steel, but here again the relatively small size of the bodies forbids the erection of expensive plant, and indicates that if these bodies are to be so used in the future, it will be by selling ore to works (probably in another State) which are based on a larger deposit of ore of similar type.

PRODUCTION.

The figures of production prior to 1903 are obviously incomplete. The records prior to 1880 are of exported material only, and take no account of the very considerable quantities that were mined as flux for the Burra and Kapunda mines, or for the numerous small reverberatory copper furnaces erected throughout the State. There is also a strong presumption that the falling off prior to 1897 is due to incomplete returns, as several plants were then at work on the smelting of Broken Hill lead ores. The period 1898 to 1902 shows no production, but the writer, in 1898, saw ironstone flux used at three smelters in the State.



Production of Iron Ore in South Australia.

Furthermore, the output of various quarries seen by the writer that were worked for flux prior to the utilisation of Iron Knob ore, is estimated from individual records and from the size of the excavations, as not less than 100,000 tons; and when the numerous smaller quarries not seen are considered, it is fairly certain that not less than 120,000 tons were mined prior to 1903, in place of the 26,537 tons of which records exist.

Production of Iron Ore in South Australia.

Year.	Tons.	Value.	Remarks.
		£	
1850.....	117	1,250	Export figures only. Flux for copper smelters not recorded.
1860.....	5	50	
1862.....	9	25	
1865.....	20	100	
1874.....	6	20	
1879.....	293	812	
1880.....	2	8	
1889.....	4,397	1,882	No record of production.
1890.....	3,967	1,108	
1891.....	6,799	4,551	
1892.....	4,184	4,183	
1893.....	3,050	3,321	
1894.....	1,564	1,592	
1895.....	1,611	1,613	
1896.....	109	109	
1897.....	404	336	
1898.....	—	—	
1902.....	—	—	
1903.....	84,932	46,712	Iron Knob ore to Port Pirie lead smelters.
1904.....	46,687	27,091	
1905.....	84,483	48,577	
1906.....	75,226	33,852	
1907.....	84,600	38,100	
1908.....	88,000	39,600	
1909.....	16,120	8,296	
1910.....	46,200	21,945	
1911.....	42,300	26,400	
1912.....	42,200	26,375	
1913.....	60,658	37,911	Steel works started at Newcastle, N.S.W.
1914.....	42,622	37,137	
1915.....	237,375	264,612	
1916.....	188,329	200,382	
1917.....	328,386	359,723	
1918.....	257,029	277,279	
1919.....	268,530	307,402	
1920.....	413,038	478,436	
1921.....	506,993	628,770	
	2,940,245	2,929,560	

THE ECONOMIC IRON MINERALS.

While iron is one of the commonest constituents of the earth's crust, the proportion being estimated at 4.44 per cent.*, the bulk of it is in a low-grade and non-available form, and natural concentration is required to make the material of economic importance.

A number of iron ore minerals may be used for the manufacture of iron, and the following list shows the chief minerals in general use, with their theoretical composition and some physical properties. In practice they are more or less contaminated by earthy impurities.

Magnetite.—Crystallisation isometric; frequently found as small octohedra, generally massive. Composition: metallic iron, 72.4 per cent.; oxygen, 27.6 per cent. Color, black to steel gray; color of streak or powder black; hardness, 5.5 to 6.5, brittle; specific gravity, 5.0 to 5.1. Magnetic, affecting the compass.

* Data of Geochemistry, U.S.G.S., pp. 33-34.

Haematite (Specular iron ore, micaceous iron ore, red hæmatite).—Crystallisation rhombohedral; massive granular, lamellar, or micaceous, or pulverulent and earthy. Composition: metallic iron, 70 per cent.; oxygen, 30 per cent. Color, iron black, steel gray, or reddish, streak or powder red to reddish brown. Hardness of crystals, 5.5 to 6.5; specific gravity, 4.5 to 5.3. Becomes magnetic on heating in inner or reducing flame of blowpipe.

Limonite or Brown Iron Ore.—This group of hydrous iron oxides is subdivided into several minerals, according to the proportion of combined water, as shown in the following table:—

	Metallic Iron.	Oxygen.	Combined Water.
	Per cent.	Per cent.	Per cent.
Turgite	66.2	28.5	5.3
Göthite	62.9	27.0	10.1
Limonite	59.8	25.7	14.5
Xanthosiderite	57.1	24.5	18.4
Limnite	52.3	22.4	25.3

Turgite.—Fibrous to massive, often botryoidal and stalactitic like limonite, also earthy. Color, reddish black to dark red, bright red when earthy; streak or powder red. Hardness, 5-6. Specific gravity, 4.29-4.49. In closed tube yields water and decrepitates.

Göthite.—Crystallisation orthorhombic. In prisms and scales, also fibrous, foliated, or massive. Color, yellowish, reddish, and blackish brown. When massive, liver brown to blackish brown and rust brown; streak or powder brownish yellow to ochre yellow. Hardness, 5. Specific gravity, 4.0-4.4.

Limonite.—In stalactitic, botryoidal, and mammillary forms, having a fibrous structure. Also concretionary, massive, or earthy. Color of fracture, varying shades of brown. When earthy, brownish yellow to ochre yellow; streak or powder, yellowish brown. Hardness, 5-5.5. Specific gravity, 3.6-4.0.

Xanthosiderite.—Fibrous or as an ochre. Color, yellow of different shades, more or less brown; sometimes reddish; streak, ochre yellow. Hardness, 2.5.

Limnite.—For the most part occurring as a recent bog iron deposit.

Siderite.—Iron carbonate. Crystallisation rhombohedral, usually massive, but sometimes concretionary. Composition, metallic iron, 48.2 per cent.; carbonic dioxide, 37.9 per cent.; oxygen, 13.9 per cent. Color, greyish white to brown, often dark brownish red, darkens on exposure; streak uncolored, translucent to nearly opaque. Hardness, 3.0-4.5. Specific gravity, 3.7-3.9. Frequently contains argillaceous material, and is then known as clay ironstone.

Pyrite and Pyrrhotite.—The sulphides of iron. These sulphides are chiefly valued for the sulphur content, but in some cases in Europe the residue left after roasting is sold to iron smelters. These minerals are of interest in any discussion of iron ores, in that they, as well as siderite, decompose either *in situ* to yield the brown iron ores, or yield soluble salts, from which concentrated deposits of iron ore are formed.

CONCLUSION.

A number of known iron localities were not visited, as they are believed to be small, but all that are known to be of any magnitude were seen. The writer believes that the most promising field for future discoveries of magni-

tude is in the counties of York and Buxton, and to the north of them, and possibly in the north-western portion of the county of Jervois, all on Eyre Peninsula.

The outstanding deposit of iron in the State is the Iron Monarch, with the adjoining Iron Knob. This is estimated to contain 133 million tons of ore in sight, above the level of the surrounding country.

Next in order of magnitude are the ore bodies of the Middleback Range. These bodies, of tabular habit, are numerous and comparatively narrow, but of great purity, and were formed by replacement of the country rock by hæmatite along crush zones.

The tonnage was estimated (on prospecting work only) as 32,000,000 tons of recoverable ore above the level of the surrounding country, of which 15,000,000 tons is within 100ft. of the surface. In view of the difficulty of measuring widths, a variation of 25 per cent. in either direction is possible on these figures.

This ore will not be available until a tramway connection with the shipping port is made.

None of the other deposits seen by the writer are likely to contain such tonnage as to be worthy of serious consideration as a basis for the foundation of a steel industry.

Only two contain more than half a million tons of available ore, and only three others contain upwards of 300,000 tons each, within 100ft. of the surface, the estimated limit of open-cut working.

Mount Bessemer, situated about 30 miles by road from Adelaide, and 22 miles direct, is estimated to contain 535,000 tons above the level of the South Para River. The ore undoubtedly persists to a greater depth, but the ore body crosses beneath the pool formed by the weir directing the South Para water into Barossa Reservoir.

The water from this reservoir serves 670,707 acres of closely settled country, and the circulation of its supply of intake water through deep mine workings is out of the question. Thus the lower limit of possible working is clearly defined. The grade is about 56 per cent. of iron with .052 of phosphorus. The deposit is only fitted in magnitude for the production of pig iron, and for the same reason the cost of bringing the ore to market will be relatively high.

Koolka, Billeroo, and the Cutana Group, all within a radius of 30 miles of Olary, on the Broken Hill railway, are estimated as capable of yielding more than 300,000 tons each.

Koolka is estimated to contain 414,000 tons, averaging 66.7 per cent. of iron and .3 per cent. of phosphorus; Billeroo, 385,000 tons, averaging 50 per cent. of iron and .6 per cent. of phosphorus; while the Cutana group is estimated capable of yielding 810,000 tons of ore containing 45 per cent. of iron and low in phosphorus. The ores of the first two are magnetite and hæmatite, while that of Cutana is essentially limonitic, with a little hæmatite.

Donnelly's, 14 miles from Quorn, is estimated to be capable of yielding 319,000 tons of limonitic ore, assaying 49 per cent. of iron and 1.60 per cent. of phosphoric anhydride.

All suffer from the cost of transport to the coast in comparison with the large deposits now being worked, and the provision of transportation facilities to a point of utilization or the seaboard would be heavy per ton of ore available.

The numerous smaller deposits described in Part II., in the absence of an active buying market for such ores, must lie idle, except in so far as they are capable of providing (1) flux to lead or copper smelters in the future as they have done in the past, (2) ochres, (3) road material.

At present there are only two deposits of the first magnitude known in the State, both controlled by the Broken Hill Proprietary Company, and containing ores of such composition that their admixture will be of mutual benefit.

Though the present condition of exchange and the high cost of production makes competition with countries of depreciated currency severe in the steel trade, the high quality, abundance, and accessibility of the raw material should, in time, be contributing factors in the building up of the iron industry to the full limits of Australian consumption.

DETAILS OF DEPOSITS

COUNTY OF ADELAIDE.

HUNDRED OF BAROSSA.

The Mount Bessemer Iron Mine.

Mineral Leases 2428, 2429, and 2430.

The writer inspected the ironstone body on November 16th, 1921, and found no work had been done to modify the report made by the Government Geologist in 1912. The writer on that occasion was present and assisted in the preparation of the plan published herewith.

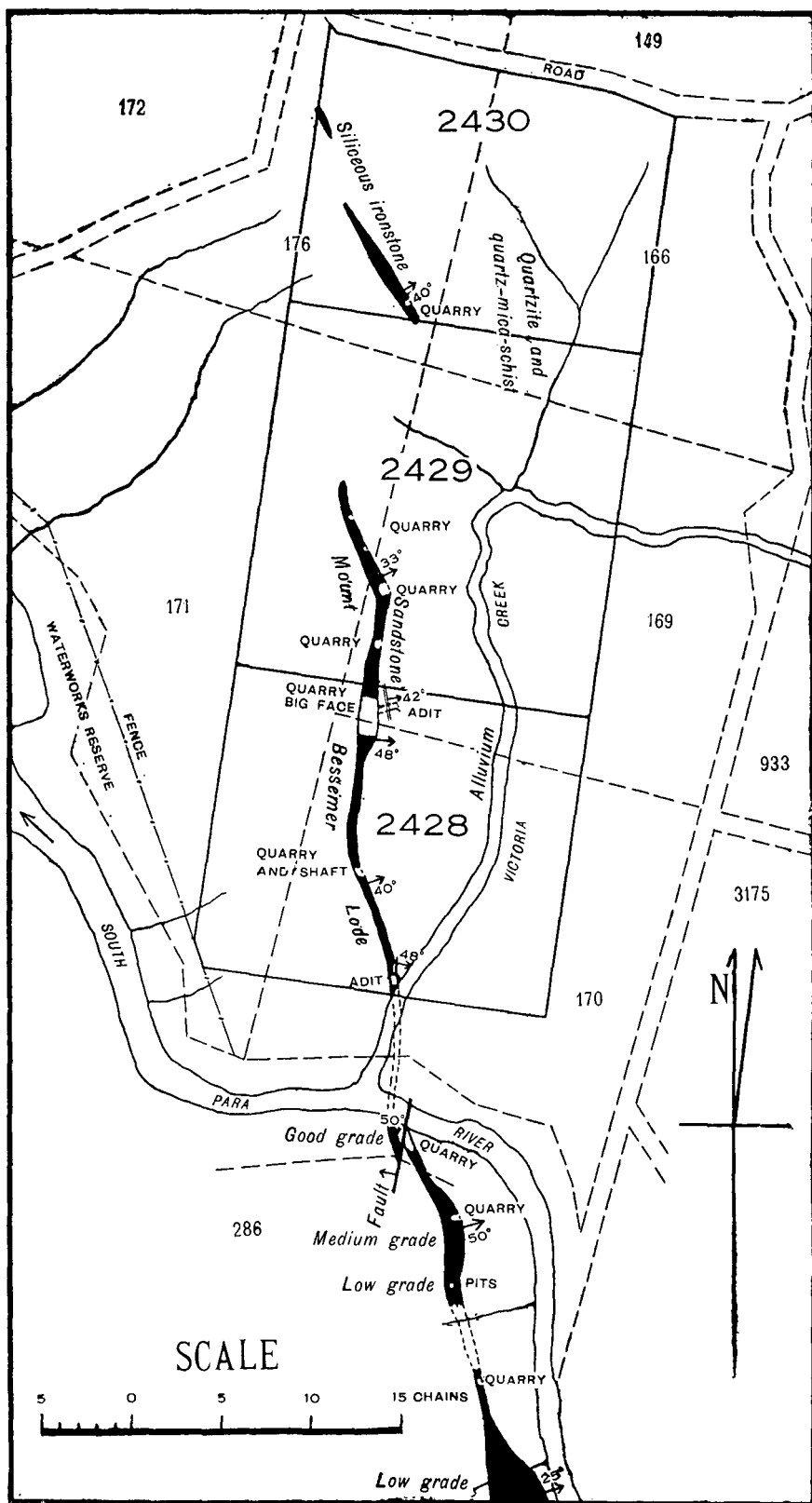
In that report the tonnage was calculated to the level of Victoria Creek only, although the lode undoubtedly extends much deeper. This limit was taken for the reason that the South Para River supplies Barossa Reservoir from a weir below the workings, and the contamination of the supply by mine water would be out of the question.

As no additional development was noted to modify the tonnage or conclusions contained in the report of the Government Geologist, his report, contained in Mining Review No. 17, is reprinted with the addition of a map prepared at that time.

Locality.

The lodes of iron ore with which this report is concerned are situated near the boundary between the hundreds of Barossa and Para Wirra, the principal lode crossing that boundary at the confluence of Victoria Creek and the South Para River.

That portion only of the lode system which lies on the north side of the South Para River is now held under mineral lease. The three mineral leases 2079, 2080, and 2081 (now 2430, 2429, and 2428, respectively) comprise an area of 108 acres, and occupy portions of the sections 166, 169, 170, 171, and 176, in the hundred of Barossa.



Plan showing the Mount Bessemer Ironstone Lode, Hundreds Barossa and Para Wirra.

History of Development.

The main lode has long been known, and has been worked from time to time as a source of iron flux. The mining for flux has been unsystematic, and a considerable number of excavations have been made at various points along the outcrop. From the largest of these openings, usually referred to as the "Big Face," some 12,000 tons of ore have been removed as far as may be judged from the dimensions of the excavation. It does not appear probable that more than 40,000 tons of iron ore have been removed from the lode over its entire length.

Some experimental smelting of the ore was attempted many years ago, and pig iron is reported to have been actually produced at Gawler. But the quantity tested was small, and no further attempts were made to utilise the ore as a source of iron. This being so, it is not a matter for surprise that no effort has been made to pierce the lode across its width at a number of places in order to provide accurate data for the estimation of quantities. The amount of work to be done to give this information is not large, as short tunnel crosscuts could be inexpensively driven on the levels of existing benches. The workings now open for observation are superficial and are located in all cases upon the hanging-wall side of the principal lode, since that portion has been most accessible, and the open cast working has hitherto involved very little stripping of overburden.

Geology.

The ore body which constitutes the main lode has certainly been formed by the more or less complete replacement of a series of tilted sedimentary rocks, with the dip of which it conforms in a general way.

The iron-bearing material is hæmatite in all cases, save where hydration at the surface has resulted in the formation of an insignificant amount of limonite. The variety of hæmatite represented is that known as specularite, and in most of the excavations the micaceous form of this variety is abundantly represented.

Intimately associated with the hæmatite are present quartz pebbles and some argillaceous material, both of which represent the undigested residues of the beds of sediment which have suffered replacement. In the majority of the exposures the undigested matter is disseminated through the iron ore; but in the tunnel on the bank of Victoria Creek, at the southern boundary of mineral lease 2081 (now M.L. 2428), there is a well-defined inclusion of country rock, *i.e.*, a "horse," the contents of which are to be discarded in mining.

Small veinlets of quartz extend outwards from the iron lode into the country rock on the hanging-wall side. These veinlets are approximately perpendicular to the plane of the lode.

The principal lode extends north for a distance of 29 chains from the southern extremity of the leases. To the north of this lode, and 9½ chains distant from its northern extremity, there lies another lode of similar material which may be traced on the surface for 7 chains. There is a still more northerly outcrop of low-grade iron ore on the western boundary of mineral lease 2079 (now M.L. 2430).

These three lodes appear to be parts of one lode system, the components of which are disposed *en echelon*.

The former of the two last-mentioned outcrops has been worked to a small extent, but the grade of ore exposed is not so high as that in the principal lode to the south.

The main lode extends beyond the limits of the area held under mineral lease, and was traced south for a distance of 30 chains from the South Para River. Beyond this point the outcrop was not followed. Very little work has been done upon this south extension of the lode, since the outcrop does not rise to such heights above the river level as it does within the mineral leases to the north, and the tonnage to be obtained from open cuts is correspondingly smaller.

The width of the lode cannot be exactly determined for lack of the necessary exposures. The outcrop varies very considerably in width on account of the comparatively low angle of dip and the relation of the ore body to the topography. Where the overburden of sedimentary rocks has not been removed by denudation the outcrop is narrow, but where extensive natural stripping of the overburden has occurred the outcrop is broad.

Existing exposures show the lode to have a width of at least 16ft., and that at some points this width is exceeded. For the purpose of calculation of tonnage it appears that the width of 16ft., measured at right angles to the dip-plane, may be reasonably taken as the average smallest dimension of the lode. It may be that future prospecting operations will extend this estimated average width with the possible addition of another 25 per cent.

The strike of the principal lode is, when the whole length is taken into consideration, 10° west of north, and that of the larger of the two ore bodies in mineral lease 2079 (now M.L. 2430) is 38° west of north. The dip is in each case to the east, and the angle of dip varies from point to point between 40° and 50° .

There has apparently been a small amount of displacement of the lode at a period subsequent to its formation, but only portions of the ore body are affected, and the general continuity of the lode is beyond doubt. There is apparently a fault plane at the north end of the Big Face, which truncates the hanging-wall side of the lode. Its strike is 10° east of north, and its dip 76° towards the east. The exposure of this fault is only a small one, but it does not appear to affect the continuity of the lode either to the north or to the south. It is possible that this fault is related to the lode channel, and that it actually antedates the period of ore formation, but further exposures are necessary before the matter can be definitely proved. A small overthrust fault, marked by the displacement of the quartz veinlets by a few inches, is visible in the hanging-wall country at this place. On the south bank of the South Para River there is a duplication of the outcrop of similar lode matter which appears to be due to a normal strike fault with a westerly dip.

The outcrop of the main lode within the Mount Bessemer mineral leases follows the east flank and crest of a ridge which extends north from the point of confluence of the South Para River and Victoria Creek. The ridge rises steeply from creek level, and the most northern portion of the outcrop attains an altitude of 150ft. above the level of the creek. The mean altitude attained by the outcrop along the face and crest of the ridge may be taken as about 100ft. above creek level. Making allowance for

the dip of the lode, and measuring the height of the outcrop along the plane of the lode in the direction of the dip, an average value of 140ft. is obtained.

From the dimensions given the cubic contents of the main ore body within the Mount Bessemer leases and above the level of the Victoria Creek may be ascertained.

It is 16 (width) x 140 (height) x 29 x 66 (length) cubic feet.

Taking 8 as the number of cubic feet of ore to the ton—

The gross tonnage above creek level = $\frac{16 \times 140 \times 29 \times 66}{8} = 535,920$ tons.

This quantity of, say, half a million tons represents that portion of the main lode which is most readily accessible for mining, being above the level of the natural surface drainage of the district.

There is no doubt in the mind of the writer but that the ore body is continuous in depth, but the deeper extension of the lode must be proved at a number of points along its course by boring before a reliable estimate of the reserves of ore at a depth can be framed. One such bore has been drilled from a point opposite the Big Face, but the exact details with respect to it are not available.

There will be a considerable amount of overburden to be removed from the lower slopes of the ridge along which the outcrop runs if an open cut is to be carried down to creek level. Below that level the mining must inevitably be underground, and special provisions must be made to cope with the water of Victoria Creek and the South Para River.

The estimates given above of the ore which lies above the creek level are concerned only with the total bulk. It is impossible, from the existing exposures, to give separate estimates of various grades of ore.

A number of samples were carefully taken at the various workings, and serve to show the composition of the ore at those points. As far as may be judged the samples represent typical ore.

The samples, of which the assay returns are given below, were taken over various widths at spots thus briefly described—

Nos. 1-4. Mouth of tunnel on bank of Victoria Creek, near south boundary of mineral lease 2081 (now 2428)—

No. 1. Footwall side of exposure. Width sampled 2ft. 6in.

No. 2. Lode-matter adjoining No. 1. Width sampled 3ft.

No. 3. Central portion of lode. Width sampled 5ft.

No. 4. Hanging-wall side of lode. Width sampled 3ft.

No. 5. Open cut on hanging-wall side of lode in mineral lease 2080 (now 2429). Width sampled 10ft.

No. 6. Open cut in central portion of mineral lease 2081 (now 2428). Width sampled 10ft.

Nos. 7-9. Big Face, in north portion of mineral lease 2081 (now 2428)—

No. 7. Footwall exposed at north end of Big Face. Width sampled 6ft.

No. 8. Hanging-wall side of lode at same place. Width sampled 7ft.

No. 9. Short tunnel in footwall, centre of Big Face. Width sampled 10ft.

Analyses by Mr. W. S. Chapman, Departmental Analyst.

	Number of Sample.					
	1	2	3	4	5	6
Iron	60.3	54.5	51.9	40.8	53.3	56.4
Silica	9.3	16.4	20.7	37.1	18.3	13.4
Phosphorus	0.04	0.07	0.04	0.04	0.07	0.06
Gold	nil	nil	nil	nil	nil	nil

		Number of Sample.		
		7	8	9
	Water at 100° C. (H ₂ O).....	0.04	0.05	0.07
	Water above 100° C. (H ₂ O).....	0.28	0.33	0.43
	Silica (SiO ₂).....	9.04	11.83	29.08
	Alumina (Al ₂ O ₃)	2.37	2.81	3.28
(1)	{ Ferric oxide (Fe ₂ O ₃)	86.83	83.41	64.57
	{ Ferrous oxide (FeO)	0.50	0.92	1.00
	Lime (CaO).....	0.02	0.04	0.08
	Titanic dioxide (TiO ₂).....	0.37	0.46	1.11
(2)	Phosphoric anhydride (P ₂ O ₅)	0.07	0.06	0.20
	Sulphur (S)	0.02	0.02	0.08
	Manganese oxide (MnO).....	nil	nil	nil
	Vanadium pentoxide (V ₂ O ₅).....	nil	nil	present
	Gold (Au)	nil	nil	nil
		99.54	99.93	99.84
(1)	Containing metallic ironFe	61.17	59.10	45.93
(2)	Containing phosphorus.....P	0.03	0.03	0.09

COUNTY OF DALHOUSIE.

SECTION 27, HUNDRED OF PEKINA.

A bold ridge on this section contains a considerable quantity of low to fair grade limonite, and some small openings had been made for the supply of flux. The quality is, however, somewhat low. This ridge is on the same ferruginous zone as that passing through section 34. No samples were taken, but as the deposit is not far from occurrences of phosphate of alumina, it is probable that the limonite contains an appreciable percentage of phosphorus.

SECTION 34, HUNDRED OF PEKINA.

On this section a hill about 100ft. above the surrounding country, composed of fairly flat bedded argillaceous and arenaceous rocks of Cambrian age, is capped by a deposit of limonite. The limonitic area has a length of eight chains in a meridional direction and a width of two chains. The deposit was at one time worked for flux for Port Pirie, and several thousands of tons have been removed from scattered openings. These openings show that the ore was not uniform, varying from high grade black lustrous limonite to valueless iron-stained rock.

Only the best material was sent away, and an analysis of this class of material (black lustrous limonite) gave the following result:—

Insoluble	2.7	per cent.
Oxide of Iron.....	81.3	"
Metallic Iron	56.9	"
Water	11.2	"
Phosphoric anhydride	2.8	"

The workings show that the deposit is of secondary origin, and that it is a local enrichment along a persistent north-south zone, closely associated with the zone of phosphate of alumina described in Bulletin 7. This ferruginous zone shows small enrichments at intervals from section 60, hundred of Pekina, to section 34, and half a mile N. 10° E. from the workings on section 34 is a smaller rise containing limonite.

COUNTY OF DALY.

SECTION 350, HUNDRED OF CLINTON.

Near the southern end of this section two small groups of workings have provided flux for the Block 14 lead smelters, formerly at Port Adelaide. Only the southern group furnished any appreciable tonnage. The main working extended 200ft. east and west by about 30ft. in length and up to 15ft. in depth. Another body of limonite about 10ft. wide had been followed from the main cut for about 50ft. in a southerly direction. The limonite occurs on the main and cross joints of an argillaceous dolomitic limestone of Cambrian age, which has been much decomposed, and which has probably set free the iron by its decomposition. The limonite is superficial, and the deposit is small and by no means high grade. It was the recognition that some of the flux from this body contained a large percentage of phosphoric acid that led to the search for and the discovery of phosphate rock in this and the adjacent sections. The output of flux was probably between 2,000 and 3,000 tons, judging from the relative sizes of the excavations and the waste dumps.

COUNTY OF FLINDERS.

HUNDRED OF HUTCHISON.

The Iron Mount Mine.

(About five miles north-west from Tumby Bay.)

The following notes have been condensed from a report dated 13/3/16 by Mr. Henry Jones, Inspector of Mines. *

A large ironstone formation can be traced for some distance along the crest of a N.E.-S.W. ridge, which rises in places to 250ft. above the coastal plain. A considerable quantity of ironstone had been mined for flux in open workings along the outcrop. Iron ore of fairly good quality is exposed for more than a chain in width, but the full width has yet to be determined by further work.

At a point where the ridge is crossed by a deep watercourse large quantities of ironstone are partly buried in the soil, and, with other indications, show that the deposit is fairly extensive.

An adit, 140ft. in length, driven from near the bottom of the watercourse, exposes alternating bands of clay-slate, kaolin, and ironstone, the

* Review of Mining Operations, No. 24, p. 61.

latter in seams varying from 6in. to 15in. in thickness, with larger bunches of ironstone. The last 30ft. of the adit shows a greater proportion of ironstone. A vertical shaft, about 20ft. east of the adit and 80ft. in depth, passed through kaolin, clay-slate, and small veins of ironstone.

Seven samples taken by the Inspector of Mines are recorded below:—

No.	Locality.	Iron.	Manganese.
		%	%
1	130ft. to 140ft. E. side of adit	41·6	—
2	120ft. to 130ft. E. side of adit	34·8	0·3
3	110ft. to 120ft. E. side of adit	38·7	0·5
4	Bottom of vertical shaft	42·1	—
5	Over outcrop on hill crest	48·0	—
6	Workings on hill crest	50·5	—
7	From side of hill	44·4	—

COUNTY OF HERBERT.

SECTIONS 83, 127, AND 128, HUNDRED OF COGLIN.

Oodlawirra Flux Quarries.

(*Vide* Record of Mines, 1908, p. 332.)

Prior to the utilization of the ores of Iron Knob for fluxing purposes ferruginous flux was quarried on these sections and railed to the various smelters treating the lead ores of Broken Hill. The quarries were at that time connected with the State railway system at Oodlawirra by a 3ft. 6in. tram line, four miles in length. The output from the quarries was 18,000 tons, of a minimum grade of 51 per cent. of metallic iron, and was sold delivered for £21,101. When the Iron Knob was opened production at Oodlawirra ceased, and the ground lay idle until early in 1919, when it was taken up for ochre by the original leaseholder.

The ore won consisted of hard dark-brown to blackish limonite. Typical samples were taken by the writer from Nos. 2 and 3 quarries, and the analyses are shown below:—

	No. 2 or Main Quarry Section 128.	No. 3 or North Quarry Section 127.
	%	%
Water at 100° C.	·54	·51
Water over 100° C.	10·34	9·83
Silica.....	7·84	9·36
Alumina	6·89	5·89
Iron (metallic).....	51·52	51·98
Sulphur	·07	0·03
Phosphorus	·47	·21

W. S. CHAPMAN, Analyst.

Though the workings are strung out along a meridional line, there is no continuity between the various bodies that have been worked. There is, however, a little to the west a very pronounced silicified zone, carrying varying amounts of limonite, and crossing the strike of the slates, which are exposed to the east in Nackara Creek, at an acute angle.

The southern or No. 1 workings, on section 83, consist of several quarries scattered over an area of approximately four chains north and south and three chains east and west. This area contains several meridional and vertical combs of dark limonite, some of which is rather siliceous. The output from the various openings on this limonite has been 4,000 tons. The material between these hard limonite bodies consists of decomposed argillaceous rocks and a large proportion of yellow ochre of good color, the latter occurring near the edges of the massive limonite. In places bodies of white siliceous material are visible in the ochre, but the separation of the larger patches is effected without difficulty. A shaft 108ft. in depth has been sunk, and a crosscut exposes 22ft. of yellow ochre. There is thus every indication that a very considerable tonnage of yellow ochre can be recovered, and calcining on a working scale has shown that a good quality red can be made by calcination of the yellow ochre.

Samples of the material from the shaft (sample 3) and from calcined ochre (sample 4) were taken, and the analyses are given below:—

	Sample 3. Raw Yellow Ochre from Shaft.	Sample 4. Calcined Yellow Ochre (Red).
	%	%
Water at 100° C.	.91	1.64
Water over 100° C.	5.36	2.29
Silica.....	53.00	49.72
Alumina	4.98	4.71
Iron	24.22	28.53
Equivalent to ferric oxide.....	34.60	40.76

The calcined material came from another working, and was of slightly better quality than that from the shaft. At the same time the analysis shows that the calcination could have been more complete. The dense limonite calcines to an even richer red, and if further experiments confirm this, there is, from an ochre point of view, a very large tonnage available, not only in No. 1, but also in Nos. 2 and 3 quarries.

The central or No. 2 quarry, half a mile to the northward, on section 128, has produced 12,000 tons of dense limonite, about an equal amount of waste being removed. The two main workings have a length of 500ft. on a N. 13° E. bearing, an average width of 30ft., and a depth varying from 0 to 40ft. Ore is standing in the south face, and from surface indications appears likely to persist for several chains. The bottom of the quarry also consists of limonite. The sides of the quarry show that the lateral limits of the ore body have generally been reached, and that it is contained in much decomposed slate, some of which is arenaceous.

A sample (*vide* page 22) was taken as being approximately representative of the material mined and sent away. Parallel veins of limonite have been opened four to five chains to the westward, but the veins are only 3ft to 4ft. thick.

No. 3 quarry on section 127 has produced 2,000 tons of flux from shallow workings. These workings show the limonitic area to be about 80ft. wide, and to contain bands of good-grade limonite (*vide* analysis of ore from No. 3 quarry on page 22), with low-grade yellow ochreous

material and quartz inclusions, in an argillaceous base. From surface indications the body thins out two chains north of the quarry and becomes definitely low grade two chains south.

About 12 chains south of No. 3 quarry an opening about 40 by 20 by 30ft. deep has been made on the left bank of Nackara Creek, but the grade of the ore exposed was low. Near this an adit, 300ft. long, extends west under a siliceous limonitic outcrop, having been driven by a Terowie syndicate shortly after the discovery of Broken Hill.

Both the adit and the quarry were made on a siliceous limonitic outcrop of about an acre, the iron content of which probably does not much exceed 20 per cent.

For nearly a mile to the north towards "Cook's Blow," a limonitic quartzose "comb" is visible, but no important outcrop of good-grade ironstone is associated with it.

The workable bodies are essentially due to secondary deposition in favorable zones, where fractures, approximately parallel to the ferruginous quartzose "comb"—which itself probably marks the course of a major fault—have traversed rocks capable of providing iron or permitting a certain amount of replacement. From the nature of the ore the bodies cannot be expected to persist to any great depth, and below water level (150ft. to 200ft.) they may be expected to die out as workable bodies. However, it is probable that more than the tonnage so far produced could be won from the various faces and from the bottom of existing quarries, though at a greater expense in moving waste than formerly. Even so, the deposits cannot be regarded as large from the viewpoint of an iron smelter. Furthermore, the grade is overshadowed by many of the hematite deposits, and the percentage of phosphorus (average 0.34 per cent.) is too high to be acceptable by a steel maker producing acid-process steel. For foundry pig the phosphorus increases fluidity, though at the expense of strength.

As a source of ochre, however, the No. 1 quarry shows a very considerable quantity of the yellow type, capable of calcination to a good red. Crushing and calcining the dense limonite formerly worked in all three quarries appears to be a method of considerable promise, and should, if successful, enable very large supplies of red oxide paint to be produced.

COUNTY OF HINDMARSH.

SECTION 52, HUNDRED OF ENCOUNTER BAY.

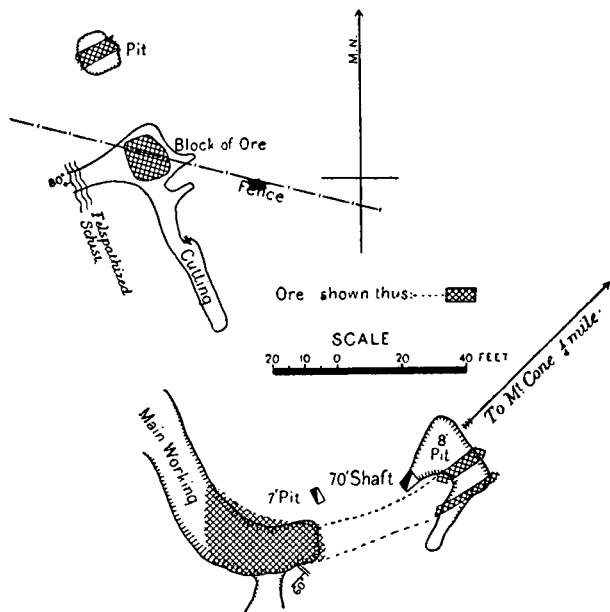
Mount Jagged Iron Mine.

(*Vide* Record of Mines of S.A., page 322.)

The workings are situated on section 52, hundred of Encounter Bay, and are approximately a quarter of a mile south-west of Mount Cone.

The country rock is felspathised schist, in which occur bunches of iron ore. Four main openings have been made, of which the relative positions are shown on the accompanying sketch. The easternmost is 8ft. in depth, and exposes a width of 12ft. of quartz and ironstone. Of this, half is iron ore carrying a small proportion of quartz. The lode here appears to strike N. 55° E., and to be nearly vertical. A shaft has been sunk in this pit, immediately to the north of the lode, to a depth of about 70ft., and the upper portion, at least, is in country. Thirty-five feet west-south-west

the main working, an open cut about 60ft. long by 15ft. deep, with its axis N.W.-S.E., is seen. Possibly the ore is continuous through the standing ground between the two cuttings.



Sketch of workings at Mount Jagged Iron Mine.

The eastern face of the main working exposes over 7ft. of ore. For 30ft. to the west the ore body is seen in the walls of the open cut, and has a width of 12ft. Some 20ft. north of the point where it is last seen in the south-west wall of the open cut, and on the opposite side of the cutting, it appears to be tapering off, and the ore carries an admixture of quartz, and is consequently of lower grade. From here to the mouth of the open cut no ore is visible. The ore broken in this working is as a whole free from gangue, and a sample taken from a heap of about 100 tons stacked in the open cut, and which may be regarded as a representative of the ore mined from this cutting, has the composition shown in the first column.

	From 100 Ton Heap in Main Working.	Second Grade Ore, Smaller Working.
	%	%
Water at 100° C.	Nil	.03
Water above 100° C.	.34	.41
Silica (SiO ₂)	.14	2.10
Alumina (Al ₂ O ₃)	.25	1.42
Ferric oxide (Fe ₂ O ₃)	87.85	86.51
Ferrous oxide (FeO)	6.09	4.88
Manganous oxide (MnO)	.06	.01
Phosphoric anhydride (P ₂ O ₅)	Nil	Nil
Titanic dioxide (TiO ₂)	5.20	4.60
Sulphur (S)	.04	.06
	99.97	100.02
Total iron (Fe)	66.23	64.35

A little northward of the main working a trench has been cut in a N. 20° W. direction along the strike of the schist. A small bunch of ore is exposed at 30ft., and at 60ft. from the south end a block of ore 15ft. x 10ft. has been cut round. This body, composed of hæmatite and limonite, with a small proportion of quartz, cannot be seen on the outer wall of the encircling trench.

Thirty feet N. 25° W. is a smaller pit, in which a body of iron and a little quartz 4ft. to 5ft. thick apparently strikes north-east and south-west. A sample from this taken as representative of the second-grade ore, has the composition shown in the second column of the above table of analyses.

Some detrital iron ore is strewn about the surface in the vicinity of these workings, but no definite outcrops were seen other than those that had been worked, and it is difficult to correlate the bodies worked, with the exception of those in the two southern openings.

The presence of 4.9 per cent. of titanite oxide (2.99 per cent. of titanium) may be noted in view of the fact that "under present furnace practice . . . the smelting of these ores is both difficult and expensive, and for that reason they are not accepted by furnaces, so that the deposits are practically worthless."*

"At present there is a prejudice, justifiable or not, against the use of ores carrying more than 1 per cent. of titanium."*

The same author points out that experiments in the direction of mixing titaniferous iron ores with titanium free ores indicate a method for their utilisation, and also quotes results which show that there is a possibility of utilising such ores by direct smelting in an electric furnace.

In places the ore of Mount Jagged contains more or less quartz.

The mine was formerly worked and the ore smelted in a charcoal furnace, and at a later period some ironstone was carted to Port Victor and shipped for flux. From the extent of the open cuts the total quantity removed can hardly have exceeded 1,000 tons to 2,000 tons.

The presence of titanium, which is regarded as detrimental in smelting, the small size of the deposit, and the fact that underground working would soon have to be resorted to, coupled with the cost of cartage, preclude the economic exploitation of this deposit in competition with larger or more favorably situated ore bodies in this State and elsewhere in Australia.

COUNTY OF HINDMARSH.

SECTION 264, HUNDRED OF GOOLWA.

Peeralilla Iron Deposit.

(*vide* Record of Mines of S.A., page 322.)

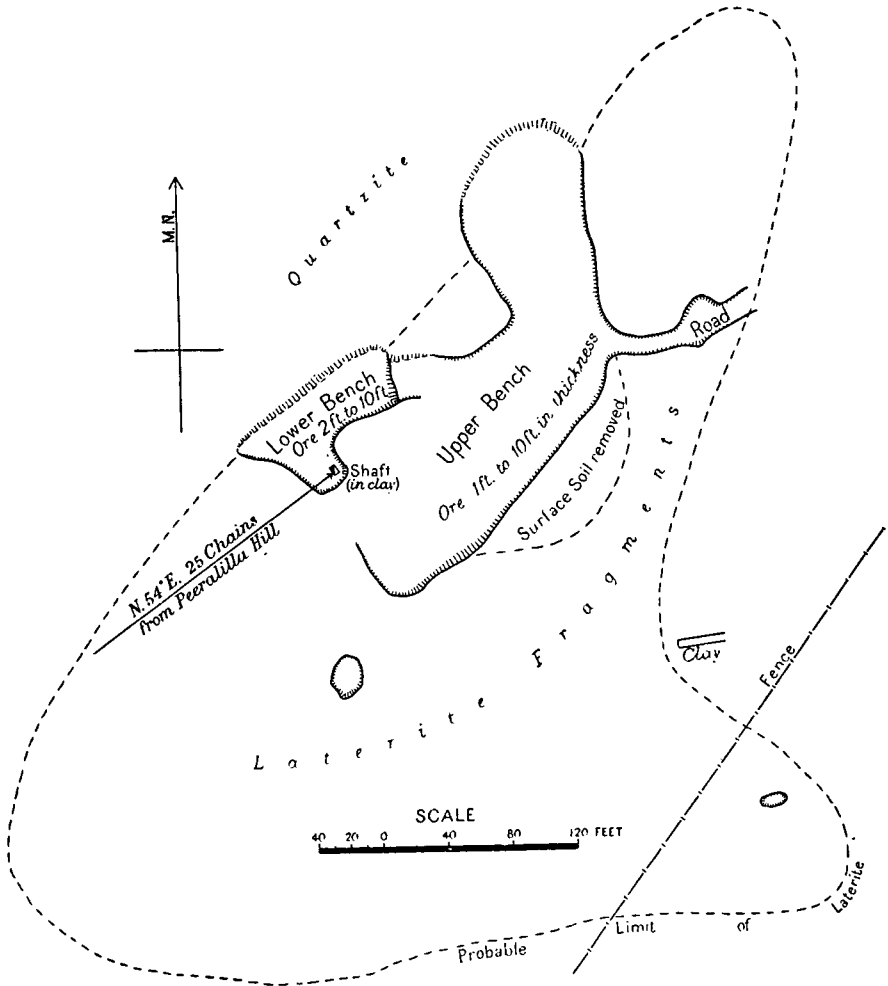
The deposit is situated 25 chains in a north-east direction from the trig-cairn of Peeralilla Hill, and is at an elevation of about 1,000ft. above sea level.

To the north-west of the working the western slope of the hill is rugged owing to the development of steeply-tilted quartzite, but the site of the workings on the crest of the ridge and the slopes to the south-east exhibit the rounded smooth topography due to the presence of argillaceous rocks.

The deposit consists of dark-brown, brown, and yellow limonite, varying in texture from earthy to massive material, and has been worked over an area of two-thirds of an acre. The upper bench of the workings show the ore to have been removed to a depth of 1ft. to 8ft., or to an

* Bulletin 64, Bureau of Mines, U.S.A., "The Titaniferous Iron Ores in the United States," J. T. Singewald.

average depth of about 3ft. An indeterminate quantity of ore is still left underfoot. At a slightly lower level a smaller bench has been taken out, and the ore left ranges from 2ft. to 10ft. in thickness.



Plan showing Peeralilla Ironstone Deposit.

A sample chipped from the face of this bench, and which may be regarded as fairly typical of the area worked, had the following composition:—

	Per cent.
Water at 100° C.	3.40
Water above 100° C.	11.90
Silica (SiO_2)	6.62
Alumina (Al_2O_3)	6.44
Ferric oxide (Fe_2O_3)	70.14
Ferrous oxide (FeO)	Nil
Magnesia (MgO)	.17
Manganous oxide (MnO)	.82
Phosphoric anhydride (P_2O_5)	.36
Titanic dioxide (TiO_2)	.08
Sulphur	.21
	100.14
Total iron (Fe)	49.10
Phosphorus	.15

A shaft has been sunk in the floor of this bench through decomposed argillaceous rocks to a depth of 20ft., and a crosscut to the south-east is visible. This prospecting was evidently done owing to a misconception of the nature of the ore body, which is purely a superficial deposit of lateritic habit.

From the area worked and the depth as shown by the sides of the excavation, it is probable that from 8,000 tons to 10,000 tons have been removed. This material was shipped at Port Victor for flux in the lead smelters.

The extent of the deposit still remaining cannot be exactly determined, as the laterite is covered by a few inches of soil, but it is improbable that its maximum extent exceeds four acres. As it is probable that the best and deepest-known portion of the deposit was worked, and as some ore is still left in the bottom of the upper bench, it might be reasonable to assume that the area unworked would yield about the same tonnage per acre as that already worked. This would indicate a possible tonnage of 45,000 tons.

COUNTY OF JERVOIS.

SECTION A1, CAMPOONA.

In 1912 the writer noted (Bulletin 3) the occurrence of a body of good-grade hæmatite in section A1, Campoona, just west of the Yadnarie-Darke's Peak road, and seven miles east of Kielpa Railway Station, on the Port Lincoln-Kimba line.

The ironstone is present as a lode 10ft. in thickness, dipping 60° to the north, with a quartzite hanging-wall and a clay-slate footwall. Prospectors had exposed the width by shallow cuttings.

A few miles to the northward, in the south-east corner of the hundred of Pascoe, hæmatite, and hæmatite-bearing quartzite lithologically identical with that to the eastward of the Iron Knob ore bodies, occur dipping steeply to the south-east, and forming low rounded hills.

No appreciable quantity of iron ore was seen here, but as the question then at issue was the underground water supply of a large area, no time was devoted to the search for possible extensions or enrichments of iron ore. The country is, however, part of that Pre-Cambrian terrain which includes the larger deposits of Iron Knob and Middleback.

COUNTY OF LIGHT.

SECTION 38, HUNDRED OF KAPUNDA.

A body of limonite is worked on this section to provide road material. It may be mentioned that the roads in the vicinity, made of this ironstone, are excellent.

The northern end is about eight chains from the northern boundary of the section, and the quarry extends for 10 chains on a bearing of S. 10° W., and has a width of 10ft. to 66ft. The widest part includes a "horse" of slate. The body dips 45° to the eastward, and is enclosed between quartzite and slate on the eastern side and slate on the footwall side.

The material quarried varies from high-grade limonite to sandstone and slate impregnated with limonite. No samples were taken, but the presence of phosphate bodies seven chains to the northward renders it highly probable that there is a very appreciable quantity of phosphorus in the ore.

COUNTY OF LIGHT.

SECTION 1738, HUNDRED OF MOOROOROO.

Limonite has been worked in the past on this section for the supply of flux to the old Block 14 lead smelters at Port Adelaide, and from the size of the openings it is estimated that about 2,000 tons have been removed from a number of small quarries. The faces show bodies of limonite ranging from 10ft. to 15ft. in thickness, and of very variable quality. A considerably greater tonnage of limonite, probably fairly high in phosphorus, could be obtained from this locality than has been obtained in the past.

COUNTY OF MANCHESTER.

Iron Knob and Iron Monarch.

M.L. 1659-1667, 2238-2242, and 2383 and 2384, all of 40 acres, except 1661, which is of 20 acres.

These leases, aggregating 620 acres, held by the Broken Hill Proprietary Company, Limited, cover not only the ore deposits, but a very considerable additional area of the plain beneath, which is necessary to include the terminal portion of the Iron Knob tramway and the connecting inclines between the quarries and the loading bins.

The leases are situated $33\frac{1}{2}$ miles west of Spencer's Gulf, and are connected with the shipping port of Whyalla by a tramway owned by the Broken Hill Proprietary Company. The tramway is 3ft. 6in. gauge, and is now laid with 80lb. rails manufactured by the company.

The annual rainfall is small, averaging $9\frac{1}{2}$ in., and the consequent lack of water has been, and is still, one of the chief natural disabilities. Hitherto small surface dams have been depended on, and when they failed, water had to be brought across the Gulf in barges and trucked to Iron Knob, or was condensed at Whyalla. A reservoir now under construction by the Government at Tassie's Creek is expected to provide a suitable supply for Iron Knob, while the company has excavated a reservoir near Whyalla to serve the town, works, and tramway.

The iron ore deposits and the associated rocks form a meridional ridge a mile and a half in length, rising to a height of about 620ft. above the surrounding country. The southern and higher portion is a large dome or boss, the central portion of which constitutes the Iron Monarch ore body or mass. The central portion of the ridge is composed of ferruginous chert or quartzite, while the backbone of the northern third is formed by the tabular ore body known as the Iron Knob.

To the southward hills continue for another mile, but in other directions the country is practically level, and, indeed, is, to some extent, built up with unconsolidated debris.

Geology.

The Knob-Monarch ridge is composed of ironstone and ferruginous cherts or quartzites, which alone make notable surface exposures. Openings show that the Knob ore body is surrounded by a laminated schist, much contorted and decomposed to below all existing openings.

On the Monarch road and other cuttings enable rather more to be seen. Along the road between Quarries A and B, and on a lower path, considerably contorted and crumpled schists and interbedded thin quartzites are visible, as a whole dipping 40° to the north-west, and forming the footwall of the Iron Monarch body.

Near the south-east end of Quarry A an abrupt change of strike comes in, and the dip is about 40° to the north-east wherever any definite structure can be found along the south-western portion of the deposit, and barren inclusions in the ore on the western slope have a similar trend. It thus appears that there must be a fault older than the ore deposition at the point of junction of the two strikes, or, at all events, an intense crumpling or folding into a pitching syncline.

The schist is decomposed at nearly every place where it is seen, but one fairly fresh exposure shows it to be chloritic in type. Pinite has developed freely in the schist at the south-eastern side of the Iron Monarch, and was noted on the north-eastern side of Quarry C. Between the A and B Quarries, and below the connecting path, an adit, made by early prospectors, follows a 2ft. vein of pinite, dipping 45° W. This green mineral is stated to have been followed in the belief that it was copper, or, according to another and more probable story, nickel.

The following analyses were made by Mr. W. S. Chapman, of samples taken by the writer in 1912, of the pinite, and of the reddish semi-decomposed schist in which the vein occurs:—

	Green Mineral, 12-5-28.	Rock Carrying Green Mineral, 12-5-28.
	%	%
Silica (SiO_2)	45.24	41.42
Alumina (Al_2O_3)	36.81	35.79
Ferric oxide (Fe_2O_3)	1.00	1.04
Ferrous oxide (FeO)	0.29	0.14
Magnesia (MgO)	0.22	0.14
Lime (CaO)	.31	0.24
Soda (Na_2O)	0.40	0.35
Potash (K_2O)	11.57	8.60
Water at 110° C. (H_2O)	0.32	0.32
Water above 110° C. (H_2O)	2.50	4.08
Carbon dioxide (CO_2)	—	0.07
Titanic dioxide (TiO_2)	1.08	6.40
Zirconium dioxide (ZrO_2)	—	nil
Phosphoric anhydride (P_2O_5)	—	0.06
Sulphur trioxide (SO_3)	—	nil
Chlorine (Cl)	—	0.01
Ferric disulphide (FeS_2)	—	0.07
Chromium sesquioxide (Cr_2O_3)	0.43	1.01
Vanadium pentoxide (V_2O_5)	—	nil
Manganous oxide (MnO)	—	0.22
Nickel, cobalt, and copper oxides (NiO , CoO , and CuO)	—	nil
Barium oxide (BaO)	—	nil
Lithium oxide (Li_2O)	nil	—
	100.17	99.96

Mr. Chapman adds the following remarks:—"From the above analysis it is seen that the green mineral is essentially a hydrous silicate of alumina and potash, corresponding in composition to muscovite, but lacking the micaceous structure and other physical characteristics of the latter, and would be classed under the general term pinite. The figures obtained agree generally with the analyses of the pinite minerals published in Dana's "System of Mineralogy," except that titanic dioxide and chromium sesquioxide are not recovered in these analyses. The presence of the former may be accounted for by the high titanium content of the rock carrying

the green mineral as shown above, and I note that more recent analyses published in U.S. Geological survey Bulletin No. 419, page 286, show chromium sesquioxide up to 0.79 per cent. The hardness (3.5) specific gravity (2.82), lustre, and color are consistent with pinité."

The predominant rock on the surface is a ferruginous siliceous or chert-like rock, which has a variable proportion of iron and a laminated structure.

The main development of this rock has a length of 60 chains and a maximum width (near the Monarch ore body) of 30 chains, and crosses the ridge between the two ore bodies diagonally, having a north-north-westerly trend. The strike of the chert is very variable, but the general dip is towards the north-east, and bodies of schist of irregular outline are enclosed in the ferruginous chert. Like the iron ore, it has resulted from replacement of the original sediments, silica being the predominant material introduced, with iron oxide as an accessory. This silicification probably slightly antedates the formation of the ore bodies, as the siliceous rock has suffered secondary fracturing and the introduction of later veinlets of hæmatite. The silica was probably derived from the deeper ground in the process of replacement, to be left higher up in the zone of deposition, or on the outskirts of the ore. Consequently there is every possibility of the north pitching body of iron ore extending for some distance beneath the chert on the north of the deposit, with a steeply-pitching hanging-wall. The junction between the hæmatite and the chert is in places ill-defined. The use of a drill on the boundary of leases 1662 and 1663 at ($\begin{smallmatrix} 120 & 121 \\ 120 & 121 \end{smallmatrix}$) about 200ft. north of the ironstone, to determine whether the chert overlies ore, would be helpful in regard to the future layout of the quarries.

Much of the hill slopes is covered by talus of hæmatite and chert. To the west the underlying rocks are masked by sand and loamy soil, and on the eastern side there is evidence of a deep depression now filled by detrital matter.

Samples from a bore sunk by the Company near the site of the present mess camp have been examined, and show the following section:—

Feet.	Feet.	
0 to	2	Travertine and loam.
2 to	21	Pebbles of ironstone, ferruginous chert, &c.
21 to	34	Travertine limestone (finely divided and soft).
34 to	70	Red brown ochreous earth.
70 to	73	Pale yellowish clay.
73 to	101	Mottled ochreous clays.
101 to	145	Pale reddish ochreous clays with quartz pebbles.
	145	Salt water, 5½oz. per gallon; 332galls. per hour over 48 hours' pumping test.
145 to	175	Coarse quartz pebbles.
175 to	180	Smaller pebbles in ochreous clay.
180 to	232	Quartz pebbles in pale ochreous clay.
232 to	260	Somewhat friable siliceous rock (like weathered quartzite).
260 to	266	White quartzite.
266 to	335	Pale clay-slate.
335 to	346	Pale arenaceous slate with black veinlets of iron or manganese oxide.
346 to	380	Pale slate, more compact.
380 to	390	Quartzite.
390 to	395	Soft friable clay slate with a few hard bands (probably quartzite).
395 to	430	Friable clay-slate with arenaceous bands.
430 to	456	No sample.
456 to	466	Felspathised mica schist.

The upper 145ft., and possibly to 232ft., is detrital filling, and the last 10ft. (456ft.-466ft.) undoubtedly belongs to the Pre-Cambrian series. The intervening material may be the lower part of the Tent Hill formation of Corunna Range. (See page 33.)

The Pre-Cambrian rocks outcrop to the south of the Monarch as a lesser hill.

Intermediate between these points, two miles south of the Monarch, decomposed gneiss or gneissic granite is to be seen on the dump of a well, which, prior to reaching the gneiss, penetrated about 40ft. of fine sand and quartz pebbles.

One and a half miles west of the Iron Monarch granite containing much felspar and biotite is visible in a small outcrop. Similar granite and an aplitic variation occurs in Burkett's Hill, eight miles north-west of the Monarch, and is known to be present for some miles to the north and west.

Four miles north of Burkett's Hill is the easternmost exposure of the great felspar-porphry mass that forms the Gawler Ranges. This felspar porphyry is probably effusive over the Pre-Cambrian surface. Its age has not been established and it cannot be definitely said that it is older than the Tent Hill formation, although pebbles of felspar porphyry are recorded in the latter.

The following analyses were made in 1912 by W. S. Chapman (Departmental Analyst) of felspar porphyry from this locality, and of granite from Burkett's Hill, for purposes of comparison with similar rocks seen in the area dealt with in Bulletin No. 1, G.S.S.A.:—

	Felspar Porphyry, No. 12-5-15.	Granite, No. 12-5-11.
	%	%
Silica (SiO_2)	68.90	63.15
Alumina (Al_2O_3)	14.43	17.68
Ferric oxide (Fe_2O_3)	1.91	2.27
Ferrous oxide (FeO)	2.80	2.43
Magnesia (MgO)	0.33	0.32
Lime (CaO)	0.92	2.37
Soda (Na_2O)	3.07	4.27
Potash (K_2O)	5.70	5.68
Water at 110°C. (H_2O)	0.29	0.10
Water over 110°C. (H_2O)	0.87	0.34
Carbon dioxide (CO_2)	0.23	0.16
Titanic dioxide (TiO_2)	0.69	0.37
Phosphoric anhydride (P_2O_5)	0.19	0.28
Chlorine (Cl)	0.03	0.02
Ferric disulphide (FeS_2)	0.15	0.09
Manganous oxide (MnO)	0.03	0.04
Strontium oxide (SrO)	—	nil
	100.54	99.57

To the south-east of the Monarch a pit opened for fireclay exposes highly decomposed crumpled schist of Pre-Cambrian age. Only slightly plastic

and non-refractory, the clay was used in conjunction with others at the Port Pirie lead smelters. It has the following composition:—

	%
Silica (SiO_2)	72.24
Alumina (Al_2O_3)	16.80
Ferric oxide (Fe_2O_3)	1.84
Magnesia (MgO)	4.48
Lime (CaO)	0.22
Soda (Na_2O)	0.14
Potash (K_2O)	3.20
Water at 100°C. (H_2O)	0.90
Water above 100°C. (H_2O)	3.02
Titanic dioxide (TiO_2)	0.76
Chlorine (Cl)	0.34
Sulphur trioxide (SO_3)	0.16
Less O = Cl	0.08
Total	100.02
Total fluxes	5.88

W. S. CHAPMAN, Analyst.

Overlying these undoubted Pre-Cambrian rocks is the extensively developed Tent Hill formation of the residual hills, so called from the resemblance, in profile, to a marquee tent.

The nearest known exposure, other than the possible occurrence in the bore referred to above, is $2\frac{1}{2}$ miles north of Iron Knob, where a bed of quartzite dipping 20° to N. 35° E., outcrops, and is overlain in ascending order by thin-bedded flaggy red sandstones, quartzite, and the quartzite conglomerates of the Corunna Range. These conglomerates, several hundreds of feet in thickness, contain pebbles of iron ore and ferruginous chert identical with that exposed at Iron Knob.

These beds have been tentatively given an Ordovician age on the structural relationship of similar rocks to the Cambrian and to the fact that lithologically they have been compared by H. Y. L. Brown* with similar beds carrying Ordovician fossils at Tempe Downs, in the Northern Territory.

There is, however, a possibility that they are allied to the Port Hughes and Winulta conglomerates†, which are overlain by Upper Cambrian fossiliferous limestones. If this is the case, they would indicate that a great overlap of the Upper Cambrian beds on the Pre-Cambrian foreland of Eyre Peninsula has taken place, and would definitely prove that the Iron Knob, Monarch, and Middleback Range ore bodies were formed in Pre-Cambrian time. They are certainly older than the Tent Hill series.

The Mount Young and Moonaby Range conglomerates, described by the writer in Bulletin 3, G.S.S.A., page 43, are probably older than the Tent Hill series. They are strongly suggestive of fluvio-glacial tillite, and may, in consequence, be of Cambrian or Upper Pre-Cambrian age.

The Ore Bodies.

The first body to be worked was the Iron Knob proper, at the north end of the iron-bearing ridge. Until 1914 the whole output, amounting to 714,028 tons, and averaging 68.5 per cent. of iron, was used as flux at the lead smelters of the Broken Hill Proprietary Company at Port Pirie, but subsequently the Iron Monarch body began to produce ore for the steel works at Newcastle.

* Record of Mines, 4th Ed., 1908, page 2.

† Bulletin 6, G.S.S.A., pages 30-31.

The total recorded output of the State to the end of 1921 is 2,940,245 tons*, and the production for the State, shown as a table and graph on page 11, from 1903 to 1921, is wholly from these deposits, and amounts to 2,913,708 tons.

The Iron Knob ore body is tabular in form, and has an outcrop length of 26 chains. There is a very slight underlie to the eastward at the north end. At the south end the underlie is more pronounced, being about 60°. Two quarries have been opened on it.

The northern or larger one, now known as C Quarry, is worked, but the southern D Quarry is no longer in use, as the ore body pinched to the northward.

C Quarry, started at 750ft. above sea level, has been opened for a distance of 570ft. in a southerly direction, and has a mean width of 180ft., and a maximum of 240ft. The face, when passing the peak, had a maximum height of over 100ft., but the height and width are both decreasing as the body is followed southwards. South of the present face the lode bifurcates, the western body extending to D Quarry. The eastern only extends a few feet, but is of considerable width. When this point is reached the present bench will be, comparatively speaking, worked out, but the floor of the quarry consists of solid hæmatite, and there will be comparatively little waste to be removed during the next lift. Bores sunk in the quarry some years ago ran out of ore into ferruginous country, after passing through hæmatite:—

No. 1, to 170ft., ore assaying 65.9 per cent. of iron.

No. 2, to 50ft., ore assaying 55.9 per cent. of iron.

No. 3, to 150ft., ore assaying 60.2 per cent. of iron.

These bores indicate a dip to the eastward, and the assay results include some low grade samples of material that could probably be separated in mining.

It is thus apparent that the lode will yield a second and third bench above the level of the tramway, and that if the size on the present bench be maintained to 150ft. in depth the wide portion of the lode, or the northern 650ft., should have in reserve $\frac{650 \times 180 \text{ width} \times 150 \text{ depth}}{8 \text{ft. to ton}}$
= 2,188,200 tons of ore, containing at least 60 per cent. of metallic iron.

South of the bifurcation the lode is considerably narrower, averaging perhaps 50ft. in width, and it is improbable that open cast working will extend more than 50ft. below the present surface, unless the special purity of this body makes it advisable to remove a larger proportion of overburden. The length of this narrower portion being approximately 1,000ft., the iron ore available is, in round numbers, three hundred thousand (300,000) tons.

The Knob lode should thus afford some 2,480,000 tons of additional ore by open-cut method. There is no doubt but that the ore body will persist to greater depths, but the ore will have to be recovered by underground mining.

The Iron Knob body is tabular in form, but the Iron Monarch to the south is not a simple vein-type, but a large replacement mass. It is bounded on the south-west by schists and quartzites dipping 40° to the north-east, and on the south-east by similar rocks dipping at about the same angle to the north-west, and thus lies in a trough pitching steeply to

* See page 12.

the north. The northern boundary is ferruginous chert, which in general has a north-easterly dip, and the ore may be expected to have a northerly pitch beneath this chert.

Inclusions of schist and ferruginous schist, having a similar dip to the north-east, are well developed in the western portion of the ore body.

A spur to the south-east of B Quarry extends through the footwall of the main mass, and may represent one of the channels through which iron-bearing solutions have been introduced.

The portion of the hill occupied by the main ore body has been intensely fractured, and so rendered permeable with the exception of the most argillaceous bands, which were incapable of maintaining open channels, and which, in consequence, have not undergone molecular replacement to the same extent as the remainder of the shattered mass. These argillaceous bands form "horses" in the ore bodies, and it is probable that they are present to the extent of approximately 9 per cent. of the whole mass (*see* page 36).

Relatively small, they may exceed a chain in width, and their removal tends to impede the quarrying of the ore unless extensive faces are developed in such a direction as to ensure that a schist band will not be exposed all along a face at any one time.

A sample of this unreplaced material (a ferruginous clay) from a band in A Quarry had the following composition:—

Water at 100° C. (H ₂ O)	1.28
Water over 100° C. (H ₂ O)	7.59
Silica (SiO ₂)	39.28
Ferric oxide (Fe ₂ O ₃)	11.34
Alumina (Al ₂ O ₃)	31.31
Lime (CaO)	0.32
Magnesia (MgO)	0.44
Titanium dioxide (TiO ₂)	2.39
Soda (Na ₂ O)	0.05
Potash (K ₂ O)	5.70
Chlorine (Cl)	0.15
Carbon dioxide (CO ₂)	nil
Total	99.85

One other inclusion of note was seen—a dyke in the northern portion of B Quarry. About 20ft. in width, it dips 80° to the south-south-west, and is strongly sheeted parallel to the strike. The dyke is completely decomposed, and shows very abundant small felspar laths altered to kaolin in a dark ground mass. The schist which it penetrated has been indurated for a few inches, and in consequence the replacement of the schist by ironstone has been incomplete. The substance of the dyke rock has also been slightly impregnated by iron, so that the dyke is older than the formation of the ore body.

Schistose structure is apparent in many portions of the ironstone deposit.

When the structure of the ore, the form of the ore bodies, the presence of ironstone pebbles in the younger conglomerates, and the great amount of denudation that has taken place since the formation of the ore bodies, are considered together, it is evident that the ore bodies now exposed were not formed at the surface, but at considerable depths, and further, that the ore will be found to extend far below the present surface, with a northerly pitch of the bottom of about 30°.

Depth of Ore Body.

Any discussion on this point soon leaves the established facts and enters the field of conjecture.

The following heights above sea level are given as of interest in this connection:—

	Feet.
Summit of Monarch	1,154
The lowest ore visible in place on the western slope	760
The footwall dipping north-east at Quarry A incline	880
The lowest ironstone exposed in the spur east of Quarry B	750
The footwall dipping north-west south of Quarry B	1,006
The bottom of a shaft in ore at Quarry B	870
The bottom of a drill-hole near Quarry A	540
The plain at the tramway	530
The magnetite outcrop, lowest portion	694

The above table shows that there is, in natural exposures, a vertical range of 394ft. in the main body, while the adjacent magnetite body is 460ft. below the crest.

The recent diamond-drill hole shows that the ore persists to the level of the plain, and this at a point where the observed dip of the footwall of 40° north-east indicated that the ore would give way to country at approximately this depth.

Assuming that north-easterly and north-westerly dips of 40° govern the lower limit of ore deposition, then so long as these dips hold good, there will be an increase of depth of 84ft. for each 140ft. north from the top of No. 3 incline. In other words, the trough of the pitching shoot should reach sea level at 1,500ft. north of the top of No. 3 incline.

Of more immediate interest is the question of the visible reserves of ore, and these may best be taken out in the following sections:—

- (a) Summit to 900ft. contour.
- (b) 900ft. contour to 750ft., 10ft. below lowest exposed ore of the main body.
- (c) 760ft. to 694ft. the lowest ore exposed.
- (d) 694ft. to 540ft. the level of the ore exposed at the bottom of the drill hole.

The area occupied by the ore body and included zones of waste amounts to 68.22 acres, and by the subsidiary magnetite body, about 1 acre.*

Of the major outcrop, 29.57 acres are practically solid exposures of iron ore, inclusive of 2.64 acres of waste zones.

The waste is thus 8.9 per cent., or, say, 9 per cent. of this area, and this percentage has been assumed to hold for such portions of the body as are masked by limonite and superficial debris of iron ore. In other words, 9 per cent. of the cubic content is allowed for waste. The areas of the contours at 25ft. intervals were taken out and are shown below. At the south-western and south-eastern sides of the deposit the footwall was assumed to dip 40° under the body, while the limits of form on the rest of the periphery were assumed to be vertical. This is on the safe side, as there is every likelihood of the ore pitching beneath the chert to the north. There, would, however, be the removal of a very hard and rapidly-increasing

*All areas were taken by planimeter from a plan prepared by the writer.

overburden to be faced in this direction, and so this probable extension of ore is not taken into account. The spur east of B Quarry is assumed to be vertical and to maintain its size. No deduction has been made for mangiferous fines, which ultimately will doubtless be utilised in conjunction with the hæmatite and magnetite of the western slope and of lower benches.

Table of Areas of Contours.

Feet.	Sq. ft.	Feet.	Sq. ft.
1,150	57,000	925	1,976,500
1,125	410,200	900	2,089,500
1,100	760,000	875	2,339,200
1,075	1,061,800	850	2,381,100
1,050	1,297,000	825	2,397,700
1,025	1,473,600	800	2,395,600
1,000	1,586,800	775	2,392,500
975	1,768,500	750	2,369,200
950	1,886,100		

From these areas the cubic content has been worked out as:—

1154ft. to 900ft.	350,353,167 cubic feet.
900ft. to 750ft.	354,415,000 cubic feet.

The specific gravity of some of the hard ore has been determined as 4.62. Taking 4.5 as a general average to allow for impurity and jointing, a long ton may be taken to occupy 8 cubic feet *in situ*.

Deducting 9 per cent. for waste and taking 8 cubic feet to the ton, the tonnage works out at:—

Above 900ft. contour	39,852,673
Between 900ft. contour and 750ft. contour	40,314,706
Between 750ft. and 694ft.	14,000,000
Between 694ft. and 540ft.	36,000,000

Total 130,167,379 tons

for the Iron Monarch deposit above 540ft., and the additional tonnage per foot depth of extension below that should be approximately a quarter of a million.

The magnetite body, occupying an acre, may reasonably be assumed workable to 100ft. in depth, giving 544,500 tons.

The ore reserves that are available without undue overburden may be summarised as follows:—

	Tons.
The Iron Knob lode, C Quarry end	2,188,200
The Iron Knob lode, D Quarry end	300,000
Iron Monarch above 900ft. contour	39,852,673
Iron Monarch, 900ft. to 750ft. contour	40,314,706
Iron Monarch, 750ft. to 540ft. contour	50,000,000
Magnetite ore body on lease 1662	544,500
	133,200,079

or, in round numbers, 133,000,000 tons above the level of the plain in the case of the Iron Monarch and within the scope of open-cut work in the case of the smaller bodies.

The writer has no doubt but that the ore will extend to greater depths, but in the absence of any development work no estimate of tonnage can be formed.

Ore Types.

The predominant ore in both the Iron Monarch and the Iron Knob deposits is hæmatite, which, when pure, contains 70 per cent. of metallic iron. For the most part it is massive and dense, and in places shows schistose structure. Some is earthy and pulverulent, and some, containing a proportion of manganese dioxide, is blue-black and exceedingly dense.

In the Iron Monarch magnetite is present. The writer was informed that near the summit the compass is deflected by magnetite. This also occurs on portions of the north-western slope, and a low-lying outcrop to the north-west is highly magnetic.

Associated with these, the primary and important ores, are occurrences of secondary and superficial limonite.

It occurs in two main type localities, and the deductions to be drawn from its presence are not necessarily identical. Where erosion is active, as on the slopes of the hill, the hæmatite is visible to the surface, but on the plateau-like crest, where drainage is comparatively inactive and soil is present, there is formed a crust of nodular and massive limonite.

On some of the slopes where schist is present beneath the soil there is a marked segregation of limonite into a superficial crust. This is especially notable on the road cut in the hillside on mineral lease 1662. In places the limonite may be seen penetrating the jointing and crevices of the schist for a few inches, but on the slopes it is essentially superficial in habit. On the plateau, however, though there are doubtless bands of schist concealed beneath the soil and limonite, conditions of drainage and retention of iron-laden moisture by the soil are such that limonite is abundantly formed, and may mask not only the schist bands, but the hæmatite itself. The costeans on this plateau, although yielding high-grade limonite, have not been deep enough to definitely prove whether the limonite is simply a widespread sheet irrespective of the underlying material, or whether, as in the cases of the Iron Baron and the western slope of mineral lease 1662, it indicates the presence of schist. Deeper costeaning from north-east to south-west is advisable to clear up this point, which has a bearing on the future extension of the quarries.

Many samples were formerly taken from these shallow costeans, in which all the ore exposed is essentially limonite, and it is probable that the results are lower than if the trenches had reached the hæmatite.

Two accessory minerals are present in the soil and superficial portions of the deposits—gypsum and carbonate of lime. Inappreciable in amount, they are of interest by reason of their occurrence.

The gypsum forms as small crystals and aggregates of crystals in the soil and crevices, while the carbonate of lime forms a travertinous coating on the ironstone blocks. Some, seen on the very peak of the Iron Knob prior to its removal, had the appearance of masses of travertine until lifted or broken.

In Bulletin No. 8, G.S.S.A., page 89, the writer pointed out that the gypsum associated with cyclic salt was considerable in amount, and, with the climatic conditions in the more arid portions of the State, was to a large extent retained in the soil. The same source may be ascribed in part to the carbonate of lime, which is found in the soil as a travertine layer, and in the iron ore in the joints and cleavages from just below the surface to a depth of 20ft. to 25ft.

Manganese.

The Iron Knob ore body is very free from manganese, but portions of the Iron Monarch deposit are highly manganiferous. The maximum development is in the vicinity of B Quarry and the spur to the eastward. Up to 17 per cent. of metallic manganese is recorded in some places, though the average of B Quarry is considerably lower. On the outcrop above B Quarry there are considerable areas in which the manganese exceeds 1 per cent., and these areas are shown on the accompanying plan. They are based on samples of ore from outcrops or shallow trenches.

The highest grade manganese ore in lumps is shipped to Newcastle and used in the open hearth furnaces, but a considerable tonnage of high-manganese fines is accumulating on a dump.

The low-manganese ore is blended with the non-manganiferous hæmatite to bring the amount to such a percentage as will not affect the working of the furnaces or raise the manganese tenor of the ore above 1 per cent.

Manganese being deposited somewhat less readily than iron, it is reasonable to expect that the upper portion of the deposit contains more than the average. At the same time, B Quarry in places is highly manganiferous to floor level, and the manganese extends down the eastern spur below floor level.

The deepest portion of A Quarry at a lower horizon and the shaft below the floor of B Quarry, are practically non-manganiferous, and it appears probable that boring will show that the ultimate solution of the difficulty due to manganese will be the opening of quarries lower down the slopes, the production of which, blended with the manganiferous ore, will reduce the proportion to the desired limit.

Although there is a considerable quantity of manganiferous ore on the east side, the size of the deposit is very great, and the ore on the western slope of the hill is not noticeably manganiferous, so that the extension of A Quarry in this direction will regulate the grade from the upper benches now being worked. At present the disposition of the manganese on the outcrop of the deposit is a serious disability in the quarrying of the ore.

The Waste Zones.

The detailed examination that it was possible to make with the aid of the subdivisional lines marked out on the ground by the Company, led to the recognition of several zones of schist, more or less ferruginous, in the ore body. It was also possible to differentiate ore outcropping in place, in which alone these waste bands could be traced, and the limonite-covered area of the plateau. Vertical sections of A Quarry also showed waste zones in the face.

Faces advancing north-easterly, parallel to the trend of the waste zones, are disadvantageously situated as a large zone may be encountered simultaneously along the face. Further the dip to the north-east locks the waste into the ore, rendering separation difficult. The best face of attack would be a stepped face from the north-east coming south-west, but the high manganese in the eastern portion of the deposit renders the opening of such a face difficult unless B Quarry be extended along the course of the dyke, past the manganiferous area into the solid hæmatite of the north-western section of the ore body. It appears that the carrying of the north-west end of A Quarry round the west side of the hill, and then working eastward, will reduce the inconvenience caused by these waste zones.

Grade.

Samples were taken by the writer in 1912 and in 1921 from several localities.

On the Iron Knob, Quarry C, the following samples were taken from the broken ore pile in the face:—

	Face, 1912.				Face, 1921.
	West 40'.	West Central 40'.	East Central 40'.	East 40'.	
	%	%	%	%	%
Water at 100° C. (H ₂ O)	Nil	Nil	Nil	Nil	.24
Water above 100° C. (H ₂ O)	0.68	0.51	0.64	0.56	.30
Silica (SiO ₂)	1.14	0.74	0.74	0.43	1.22
Alumina (Al ₂ O ₃)	0.47	0.17	0.32	0.09	2.69
(1) { Ferric oxide (Fe ₂ O ₃)	96.55	95.88	96.24	97.09	94.87
{ Ferrous oxide (FeO)	0.34	2.39	1.71	1.67	.50
(2) Phosphoric anhydride (P ₂ O ₅)	0.03	0.03	0.06	0.04	CaO .52
Sulphur	0.04	0.05	0.06	0.07	.11
Manganous oxide (MnO)	0.03	0.08	0.07	0.07	MnO ₂ .24
					Co ₂ .22
	99.28	99.85	99.84	100.02	100.97
	Less than 0.02	Less than 0.02	Less than 0.02	Less than 0.02	Nil
Titanic dioxide (TiO ₂)	Nil	Nil	Nil	Nil	—
Gold and silver (Au and Ag)	67.86	68.98	68.70	69.27	66.79
(1) Containing metallic iron (Fe)	0.01	0.01	0.03	0.02	.05
(2) Containing phosphorus (P)					

W. S. CHAPMAN, Analyst.

These analyses are closely representative of the grade of ore hitherto mined from C Quarry.

The mean of the assays from the borings below C Quarry made by the Company is 62.2 per cent., but as some very low-grade assays are included, it is probable that the grade of recoverable ore will be somewhat higher.

The Iron Monarch.

The following samples were taken as typical:—

- A. From the outcrop on M.L. 1662 and 1663 over 8 chains on an east-west line on subdivisions 76-73 (M.L. 1662) and 73-74 (M.L. 1663) (taken in 1912).
- B. Grab sample from broken ore along north-western half of face of Quarry A.
- C. Grab sample from south-eastern half of face of Quarry A.
- D. Sample of ore exposed in Quarry B north of dyke.
- E. Grab sample from broken ore in north half of that part of Quarry B south of the dyke.
- F. Grab sample of south portion of Quarry B south of sample E.
- G. Grab sample from dump of magnaniferous fines from Quarry B.

	A	B	C	D	E	F	G
	%	%	%	%	%	%	%
Water at 100° C. (H ₂ O)	Nil	·39	·48	·12	·26	·23	·27
Water above 100° C. (H ₂ O)	0·70	·97	1·79	1·48	2·16	2·07	1·69
Silica (SiO ₂)	1·00	2·06	1·28	·56	·50	1·88	1·24
Alumina (Al ₂ O ₃)	0·19	2·10	1·59	3·27	2·16	2·07	2·00
(1) { Ferric oxide (Fe ₂ O ₃) ..	96·07	86·67	90·64	91·07	84·20	77·03	80·02
{ Ferrous oxide (FeO)	1·42	—	2·05	1·44	—	—	—
Lime (CaO)	—	1·24	·72	·72	·76	·68	·64
(2) Phosphoric anhydride (P ₂ O ₅)	0·05	·09	·17	·32	·24	·15	·17
Sulphur (S)	0·06	·02	·07	·03	·04	·05	·08
Manganese oxide (MnO)	0·12	—	—	—	—	—	—
Manganese dioxide (MnO ₂)	—	6·19	·90	·82	10·12	16·09	14·27
Carbon dioxide (CO ₂)	—	·88	·20	·18	·49	1·05	·64
	99·61	100·61	99·89	100·01	100·93	101·30	101·02
	Less than						
Titanic dioxide (TiO ₂)	0·02	Nil	Nil	Nil	Nil	Nil	Nil
Gold & silver (Au & Ag)	Nil	—	—	—	—	—	—
(1) Containing metallic iron (Fe)	68·36	60·67	65·03	64·87	58·94	53·92	56·01
(2) Containing Phosphorus(P)	0·02	·04	·07	·14	·10	·06	·07

NOTE.—The ferrous oxide figures in samples C and D and the "G" Quarry, 1921, face sample include the calculated correction for the oxidising action of the manganese dioxide present. In the highly manganic samples (B, E, F, G) the state of oxidation of the iron was not determined, and the total iron is stated in terms of ferric oxide.—W.S.C.

Other samples taken by officers of the Department are shown below:—

H. Iron Monarch	} Analyses published by H. Y. L. Brown, "Record of Mines, South Australia, 1908."	} Samples taken by Chief Inspector of Mines. Analyses by W. S. Chapman, 28/6/18.
I. Iron Monarch		
J. Iron Monarch		
K. Iron Knob and Iron Monarch ore from loading belt at Whyalla		
L. Quarry A, north-west half.		
M. Quarry A, south-east half.		
N. Quarry B, south end.		
O. Quarry B, north end.		

	H	I	J	K	L	M	N	O
	%	%	%	%	%	%	%	%
Water at 100° C. (H ₂ O)	—	—	1·23 {	0·56	0·81	1·08	·41	·34
Water above 100° C. (H ₂ O)	—	—		2·09	2·99	3·83	1·35	1·64
Silica (SiO ₂)	1·2	2·3	1·19	3·16	1·00	4·38	1·96	1·22
Alumina (Al ₂ O ₃)	—	—	2·47	1·76	1·34	2·47	4·00	1·15
Ferric oxide (Fe ₂ O ₃)	—	—	92·88	89·04	91·36	85·76	87·00	82·44
Ferrous oxide (FeO)	—	—	1·36	2·58	1·34	1·00	—	—
Titanium dioxide (TiO ₂)	—	—	Nil	0·35	·39	·40	·07	·10
Phosphoric anhydride (P ₂ O ₅)	—	—	·04	0·08	·15	·08	·08	·13
Sulphur (S)	—	—	Nil	0·05	·18	·12	0·11	0·11
Manganous oxide (MnO)	—	—	—	0·20	·13	·32	—	—
Manganese dioxide (MnO ₂)	—	—	—	—	—	—	4·39	12·62
Total	—	—	—	99·87	99·69	99·44	99·37	99·75
Total iron (Fe)	54·57	52	66·08	64·34	64·99	60·81	60·90	57·71
Total phosphorus (P)	—	—	—	0·03	·06	·03	·03	·06
Manganese (Mn)	12·47	15	—	—	—	—	2·9	7·98

By the courtesy of the Company the following analyses, showing the grade of ore shipped from Whyalla during two periods, are available:—

	16th Dec., 1918— 2nd Aug., 1919. Per cent.	10th Sept., 1919— 4th Feb., 1920. Per cent.
Insoluble	2.61	2.14
Iron	64.7	64.4
Manganese	.41	.46
Phosphorus	.047	.039

All the quarries having contributed to the shipments, the analyses must be very nearly an average of the ore bodies.

C Quarry ore body being slightly higher grade, the Monarch deposit probably contains between 63 per cent. and 64 per cent. of iron.

As manganiferous fines are being dumped from B Quarry, the manganese content of the iron ore of the Monarch is probably a little higher than shown in the analyses.

Methods.

The ore is worked open cast in three main quarries, and no overburden is present. There are, however, certain bands of waste rock which are, as far as possible, broken separately and placed in waste dumps.

Drilling is done by hammer drills, using hollow steel, and driven by compressed air from a central power-plant serving the three quarries.

The broken materials are loaded by hand or by two steam shovels into five-ton bottom-discharge trucks, which are lowered by gravity planes to the storage bins beside the tramway.

From these bins the ore is taken in 15-ton hopper trucks to Whyalla, and after crushing, is dumped to await shipment.

The storage dump, capable of containing 30,000 tons, of which half is "live" ore, extends over the first of a series of conveyor belts, on to which the ore is led by chutes, and taken along a jetty for a distance of 2,400ft. to be discharged into vessels of 5,000 to 6,000 tons capacity.

The output at the present time is about 15,000 tons a week, and this will be readily increased as the demand warrants.

The township of Whyalla is the local headquarters of the company, and a large modern power-house furnishes power for the crushing and transporting of the ore, the lighting of the town, and a large fitting shop. In addition there is plant for the distillation of sea water for engine and domestic use.

In conjunction with these works, numerous cottages have been erected by the company for the housing of their employes at Whyalla and Iron Knob. The older ones are of wood and iron, but the more recent are built of reinforced concrete. In both townships a boarding-house, or "mess camp," has been built by the company for the convenience of the single men. The latter are housed either in tents or in the series of reinforced concrete quarters which are superseding the former.

Two large and modern institute buildings, with large hall, reading, writing, billiard and games rooms, have been recently completed.

Conclusion.

The ore bodies of the Iron Knob and Iron Monarch are deep-seated in their origin, and are likely to persist to a very considerable depth. Below 750ft. R.L. the ore is proved to extend another 210ft. in depth by one bore-hole, and this at a point where the observed dip of the footwall or bottom of the deposit indicated that bottom would be reached at about this depth.

To the north of the borehole the ore may be expected to persist to a considerably greater depth. On calculating the tonnages to R.L. 540ft. for the main body, and to depths easily attainable in the lesser bodies, ore to the extent of 133,000,000 tons can be safely estimated as in sight. To this level there should be but little waste to be removed from the boundaries of the ore bodies, but a proportion of waste, estimated at almost 10 per cent. in bulk, additional to the iron ore tonnage, will have to be disposed of.

Grade.

The gangue minerals average less than 5 per cent., the balance of the ore consisting of iron and manganese oxides.

The manganiferous ore is somewhat irregularly distributed over the outcrop, and in places has a considerable vertical range (*e.g.*, the height of B Quarry). The ore highest in manganese is useful for special purposes, but the bulk requires to be blended to reduce the manganese content. The presence of these manganiferous areas to some extent determines the position of the faces, and will cause difficulty in keeping the manganese within the limit allowed, so long as the first "lift" is being worked. The probability is that the lower portion of the deposit will be less manganiferous. Should boring establish this, a face at a lower horizon would give an opportunity to work in the excess manganese of the outcrop. The utilization of Middleback Range ore will do the same, as it is high grade and free from manganese.

In view of the large tonnage of manganiferous ore and the difficulty of keeping the manganese in the ore sent to Newcastle below certain limits, the question of its utilization is important.

Some could probably be used as flux in the lead smelters at Port Pirie, but the making of a manganiferous pig iron will probably be the solution of the difficulty. At present the high manganese patches are so placed as to make the direction of advance of the quarry faces liable to run into bars of waste from the most awkward angle. When the first "lift" is taken off the hill—which may not be for twenty years or more—it is unlikely that there will be any excess of manganese. Indeed, it is probable that it should be conserved by opening quarries lower down the hill to obtain manganese-free iron to blend with the manganiferous ore.

The value of these iron deposits is very great, and though mention has been made of certain disabilities, the disabilities are, when compared with those of other deposits in the world, trifling, especially when the very high percentage of iron in the ore is taken into account. The deposits support a considerable population in South Australia, and a still greater one in Newcastle, where not only the steel works, but an ever-increasing group of subsidiary industries essential to the wellbeing of Australia is established.

The possession of two such exceptionally rich and large ore deposits as those of the Iron Knob and Middleback ensures to South Australia for at least a century a commanding place in the production of iron ore in Australia.

While it is to be regretted that conditions governed by the cost of assembling ore, coal, and flux, and the disposal of the product most economically, placed the steel works beyond the State's boundary, it must be recognised that it is better to supply raw material to be transformed into useful products as economically as possible, than to attempt to carry on an industry at a site where the natural conditions are less favorable, and where the cost of the product would be increased.

COUNTY OF NEWCASTLE.

SECTION F, HUNDRED OF YARRAH.

Donnelly's Ironstone Quarries.

(Also known as the Comstock Mine.)

(Vide Record of Mines, page 333.)

The deposits are on lands the mineral rights of which are vested in the Crown, and are situated 14 miles from Quorn, which is 25 miles by rail from Port Augusta.

The ironstone was at one time used for flux, 17,500 tons having been shipped to Port Pirie.

The deposits are in the valley of a small creek flowing north into Mount Arden Creek, near the south-east corner of section 147. The valley is bounded on either side by massive quartzite beds between which argillaceous rocks occupy the basin.

The ore bodies occur on either side of the valley, and may be summarised as the north-eastern, north-western, south-eastern or main quarry, and the south-western groups.

The ore consists for the most part of limonite, with a little hæmatite, and has more or less completely replaced some beds near their outcrops. In several places gullies tributary to the main creek cross the limonitic outcrops, and show that the bodies, which for the most part occupy the higher ground, are superficial in habit and do not persist in depth. This is confirmed by an adit in the south-western group, which passes beneath a limonite outcrop, but only penetrates decomposed argillaceous rock.

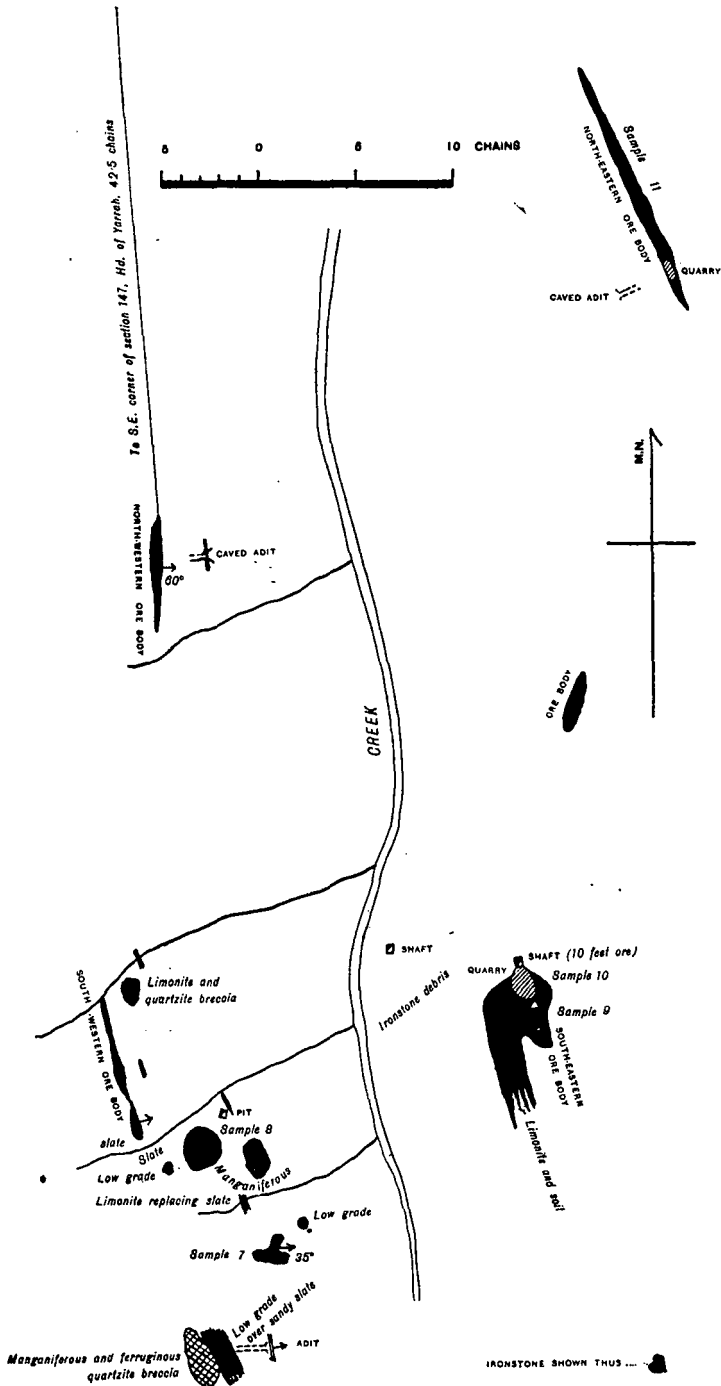
Other workings show by the absence of ore on their dumps that negative results were obtained in them. In considering the tonnage that probably exists, this factor of non-persistence in depth is taken into account for each group. The bulk of a ton of ore in place is taken at 10 to 12 cubic feet, the latter figure allowing for a proportion of separable impurity where such procedure would be advisable.

The Deposits in Detail.

The north-eastern deposit has had a working approximately 30 x 70 x 30ft. high made upon it, and consists of good quality limonite (*see* analysis No. 11 on page 47). The body has a length of 700ft. and an average width of 40ft. As to depth, 30ft. is showing in the quarry, with ore underfoot, and at the north end, and it seems reasonable to assume an average depth of 50ft., and, in view of the solidity of the body, a factor of 10 cubic feet to the ton, giving the following tonnage:—

$$\frac{700 \times 40 \times 50}{10} = 140,000 \text{ tons.}$$

The south-eastern or main quarry body has an area of about 40,000 square feet of limonite exposed and about 40ft. of a face is visible in the highest place. In view of the configuration of the knoll forming the deposit, and the evidence of a shaft immediately to the north, which passes through the limonite a few feet below the surface, it would be inadvisable to count on a greater mean depth than 30ft. A factor of 12 cubic feet to the ton is assumed, as there is a proportion of gangue visible.



The tonnage is then $\frac{40,000 \times 30}{12} = 100,000$. A sample (No. 10) was taken from the quarry, and the analyses will be found on page 47.

Between these two workings is an exposure of iron ore forming a fairly prominent outcrop, but its tonnage was not estimated.

On the western side of the valley the north-western deposit consists of a medium-grade body visible for 400ft. in length, and having a mean width of about 16ft. The eastern side forms a low escarpment dipping 60° to the east, and a small quantity of ore has been removed from a number of small faces in the scarp. Fifty feet lower down the hill a crosscut adit is driven towards the body, but has fallen in. At the south end of the outcrop the bed of a gully 50ft. below the outcrop has been eroded below the bottom of the ore body.

The mean depth is taken at 30ft., and the density at 12ft. to the ton, giving for this body $\frac{400 \times 16 \times 30}{12} = 16,000$ tons.

Two and a half chains to the eastward, at the mouth of the adit, a parallel and smaller body about 10ft. thick is visible for a short distance on the surface.

The south-western group comprises a large number of outcrops that may be roughly divided into two types—more or less linear deposits and purely surface bodies of irregular dimensions.

The northernmost exposure of this group is a body of good-grade limonite striking north-north-west, about 2 chains in length and 10ft. wide, on the north bank of a gully. Three chains to the south-west, on the opposite bank, a parallel body of about the same width extends for 4 chains in a south-south-east direction. It then thins out and makes again as an irregular body 3 chains long and 20ft. wide. A gully at the southern end shows that the limonite gives way to slate at a depth of a few feet. A small amount of ore has been broken here and there along the outcrop.

The tonnage of this group is based on a depth of 20ft. and a density of 12ft. to the ton:—

$$\frac{530\text{ft.} \times 10 \times 20}{12} = 8,000 \text{ tons.}$$

Two other smaller bars of limonite are visible to the eastward. East of and south-east of the bodies containing 8,000 tons above described are a large number of patches of limonite, some highly manganiferous (*see* sample No. 8, page 47), and for the most part obviously very shallow. The southernmost is definitely proved to be superficial by an adit beneath it, and gullies show the others to be of small vertical extent. (Sample No. 7, page 47.)

The aggregate area of these superficial outcrops is about 55,000 square feet, which at 10ft. mean thickness, and an average density of 10ft. to the ton, would be equivalent to 55,000 tons.

To the westward of the adit of the south-western group is an outcrop, one by three chains, of low-grade iron and manganese ore associated with quartzite and chert breccia, and having the appearance of a silicified breccia, cemented by limonite and a little hæmatite. The body was too low-grade to include in the above estimate of tonnage.

The grade of the ore is shown by the following samples. Nos. 1 to 6 are from the Record of Mines, and are samples taken by Mr. H. Y. L.

Brown, late Government Geologist. Nos. 7-11 were taken by the writer from the following localities:—

No. 7. Outcrop 2 chains north of adit, south-west group.

No. 8. Manganiferous outcrop, south-west group.

No. 9. Peak above main quarry or south-east workings.

No. 10. Main quarry or south-east workings.

No. 11. North-east workings.

	1	2	3	4	5	6	7	8	9	10	11
	Ferruginous Manganese Ore.	Siliceous Hematite.	Limonite.	Limonite.	Siliceous Hematite.	Siliceous Limonite.	Siliceous Limonite and Manganese Ore.	Ferruginous Manganese Ore.	Limonite.	Limonite.	Hematite and Limonite.
	%	%	%	%	%	%	%	%	%	%	%
Water	—	4.61	9.06	11.22	5.04	8.54	6.64	6.38	10.98	11.14	8.70
Insoluble, silica, and silicate of alumina	1.69	12.79	1.39	0.67	26.10	20.29	32.71	14.08	11.38	9.29	15.78
* { Ferric oxide ..	32.96	74.95	79.75	83.84	63.68	62.71	46.00	24.60	70.93	73.83	69.08
{ Ferrous oxide .	—	2.52	1.94	Trace	Trace	2.23	—	Not determined	—	—	—
Alumina	8.43	3.99	6.30	0.89	4.08	4.74	7.84	2.23	3.81	3.77	5.57
Phosphoric acid .	1.22	0.93	1.34	2.35	1.16	1.00	1.07	0.93	2.69	2.59	1.21
Sulphur	0.11	0.06	0.18	0.05	nil	nil	—	Not determined	—	—	—
Titanic acid	nil	nil	nil	nil	nil	nil	0.34	0.24	0.15	0.11	0.30
† Manganese dioxide	54.57	—	—	—	—	—	—	49.03	—	—	—
Copper	—	—	—	—	0.16	—	—	—	—	—	—
Lime	—	—	—	—	—	—	1.36	2.04	nil	nil	0.22
	98.98	99.85	99.96	99.02	100.02	99.51	—	99.53	99.94	100.73	100.86
* Containing iron .	23.07	54.43	57.34	58.68	44.57	45.63	32.20	17.22	49.65	51.68	48.36
† Containing man- ganese	34.50	—	—	—	—	—	—	30.99	—	—	—

To summarise, the bodies consist essentially of limonite, and are likely to cut out at shallow depths.

In estimating tonnage for each group consideration was given to this, and a depth was taken to which the bodies may reasonably be expected to persist.

The probable quantity of ore available is set out below:—

	Tons.
North-eastern workings	140,000
South-eastern workings	100,000
North-western workings	16,000
South-western workings	63,000
Total	319,000

To this should be added a proportion for the body between the north-east and south-east quarries and for certain possible extensions beyond the limits of ore bodies taken into account.

It would thus appear reasonable to regard a total quantity of 350,000 tons of ore as being obtainable.

The grade is to some extent a matter of selection of ore. Excluding samples 1 and 8, which are highly manganiferous, the mean of nine samples is:—

Water	9.44 per cent.
Insoluble	14.71
Iron	49.17
Phosphoric acid	1.60
Sulphur	.06

It is probable that if the bodies were worked as a whole this grade could reasonably be expected.

COUNTY OF YORK.

MINERAL LEASES 2386, 2387, 2388, 2389, 2390, 2391, 2392, 2397, 2398, 2399, 2400, 2401, 2402, AND 2403.

Iron Prince and Iron Baron Leases.

MIDDLEBACK RANGE.

The Broken Hill Proprietary Company holds the leases enumerated above (totalling 560 acres) on an iron-bearing series of ridges, which form the northernmost extension of the Middleback Range North. Situated 17 miles due south of Iron Knob, and 14 miles from the best point of connection with the Whyalla-Iron Knob tramway, they are not in a position to produce ore until a tramway is built. Consequently the work being carried out is of a preliminary nature, such as the cutting of roads over the hills, and the provision of a preliminary water supply. The roads, being mostly sidecuts, are very effective in prospecting the deposits, and, unlike costeans, have a permanent value. A considerably greater amount of information must yet be obtained before the best localities and methods for opening the quarries and for shipping bins and tramway location can be determined.

The northern group of leases is known as the Iron Prince, and the southern group as the Iron Baron.

Accuracy of Examination.

The leases, roads, and several traverse lines on the ridges have been surveyed by the Company.

These lines were run by the writer and the geological features and position of ore bodies noted during paced traverses and plotted. Aneroid readings were taken in considerable number, and the error due to variation in pressure determined by closing on to starting points, or points of which the elevation had been determined by previous traverses. The error was then distributed over the traverse. The closures both for distance and height are reasonably close. The greatest difficulty was the determination of widths of ore bodies, owing to the fact that the surface is covered with hæmatite debris. For this reason the occurrence of any particle of country rock in the debris was taken as evidence that the underlying material was not ore. Despite this it is possible that the widths of bodies are over-estimated. On the other hand, slaty debris may work on to an outcrop of ore and smaller bodies of ore may not be recognised. The writer therefore considers that an allowance of ± 25 per cent. on the estimated area of outcrops is justified in the present state of exposures, and that for closer estimation it would be necessary to subdivide the ore-bearing areas into small blocks to be examined in great detail with the cutting of costeans and trial holes, to define closely the boundaries of the ore bodies.

As, however, the estimated area of hæmatite outcrops is 31 acres, in bodies averaging 30ft. or more wide, equivalent, at 9 cubic feet to the ton, to 14,890,000 tons within 100ft. of the surface, there is a considerable tonnage even if the estimate is in excess by 25 per cent.

Topography.

The hills in which the ore bodies occur have a maximum elevation of a little over 500ft. above the plain which extends to the west, north, and east of the leases, and which is between 600ft. and 700ft. above sea level.

Geology.

The hills containing the ore bodies are made up of rock of schistose or slaty structure, very much altered by decomposition, and by siliceous zones and bands or beds of intercalated hæmatite and siliceous material. This latter rock will be referred to as ferruginous chert. The remainder of the ridge consists of hæmatite in veins and replacement masses. The relationship of these rocks to one another is set out approximately on the last plan.

The similarity of these deposits in their geological aspects to the Iron Knob and Iron Monarch deposits is very striking, and there is no room for doubt but that they are identical in type, in time of formation, in the character of the ores, and of the secondary rocks formed during the deposition of the ore. The Iron Monarch replacement, however, has been much more complete than in the case of the Middleback deposits, where the country rock between the ore-filled fissures remains, and has not been almost wholly replaced by hæmatite. The Middleback deposits, therefore, have special interest in that they afford evidence as to the origin of both deposits.

The oldest rocks of the district are slaty and schistose rocks, so much decomposed wherever exposed that no very definite idea of their original appearance can be formed. They form the core of the Middleback Range, and enclose the iron-ore bodies, which, together with the ferruginous cherts, have, by their resistance to degradation, formed the range as a residual block. On either hand they are flanked by comparatively small developments of ferruginous chert, and some of the ore bodies gradually grade into this material as they are followed along their strike.

The general strike is north by east to north-north-east, and there is a transition along the line of strike from the argillaceous schist into quartzite schist. On either flank of the range the schist merges into gneiss.

The plains on either side of the range are covered by sand, which has blown from the west through the gaps and covers the lowest saddle. The gneiss extends for some distance to the east, to the Post-Tertiary fault forming the western limit of the coastal plain*, and appears to have been penetrated by later, though still Pre-Cambrian granite, a typical example occurring at Carolby rockhole, in the Hundred of Ash.

Some miles to the eastward an isolated hill (Mount Young) rising 440ft. above the Tertiary coastal plain, consists of a tilted bed of coarse gritty quartzite, containing rounded stones of varying size irregularly dispersed through it. These stones consist of quartz, white and dark quartzite, ferruginous chert, felspar and quartz porphyries, granite, and slate. Probably of fluvio-glacial origin, the Mount Young series would thus be

* Bulletin 3, Geol. Surv. S.A., pages 40-42.

either of the Permo-Carboniferous or the Cambrian or earlier period, with a strong presumption in favor of the latter. As the included pebbles, both here and in the Moonaby Range to the south-west, are those of the Middle-back terrain, which is typically Pre-Cambrian, it follows that the ferruginous cherts, and probably the iron ore bodies, are Pre-Cambrian in age. This conclusion is strengthened by the finding of actual fragments of iron ore, in addition to the Iron Knob rocks, in the conglomerates of Corunna Range to the north of Iron Knob, and these conglomerates are at least Ordovician in age, and may even be the basal beds of an Upper Cambrian overlap on to the foreland of the Pre-Cambrian complex of Eyre Peninsula.

When the age of the formation of the iron deposits is considered, in the light that fragments have been found in the Corunna conglomerates, as well as the amount of denudation requisite to form the conglomerates and associated beds, it is apparent that the existing deposits were not formed at the surface, but at very considerable depths.

Prior to the formation of the ore bodies there was extensive fracturing in the schist, approximately parallel to the strike, but there is a tendency for the fractures to run together in plan. In places, notably on the north-eastern lease, on the road, evidence of this fracturing is afforded by brecciation of the schist in zones 20ft. to 30ft. wide.

Another lesser series of fractures, approximately at right angles, has taken place almost simultaneously, and has afforded channels for the iron-bearing solution from which it has penetrated into the meridional series of fractures, or at least into planes of weakness along the foliæ of the schist, to form a series of lenslike bodies of which the major axis cross the east-west fractures. Examples were seen on mineral leases 2388, 2400 (two), and 2401. Others may be present, but are obscured by talus debris.

The heat and pressure at the depth at which the ore bodies must have been formed stimulated the solution of the country rock and its replacement by hæmatite beside the fracture zones.

In considering the origin of the iron ore, the ferruginous chert must be considered in association with it. The following facts relating to the chert were noted either at Middleback or Iron Knob:—

- (1) It consists of interleaved cherty quartz and hæmatite in very variable proportions.
- (2) A gradual merging from pure iron ore to very siliceous chert was noted on following a vein along its strike.
- (3) At the larger development of iron—the Monarch and Iron Knob—there is a very much greater development of chert than at Middleback.
- (4) At Iron Knob, although the strike of the main body as a whole is north-north-west, the dip of the foliæ making up the ferruginous chert is about 40° to the north, and the chert is stratigraphically above the ironstone. The line of demarcation is, in places, ill defined.
- (5) On weathering, it loses its iron, a soft, easily-abraded siliceous schist being left. As this schist disintegrates very easily, it is exceptional, and generally the surface of the ferruginous chert is quite unaltered.
- (6) The chert shows minute cracks and faults, filled by hæmatite, crossing the older hæmatite in the foliæ.

The Source of the Iron.

Three sources are possible:—

- (a) From below by ascending solutions receiving their iron from relatively distant sources.
- (b) From above by descending solution, the iron being derived from material now denuded, such as—to instance the Iron Monarch—the ferruginous chert.
- (c) By lateral and descending segregation from the schists which now, in their decomposed state, contain a proportion of iron.

In favor of the theory of ascending solutions, is the fact that many of the South Australian pegmatites contain hæmatite as an original mineral, and the gneisses or pegmatized schists visible may well have been fels-pathised from a deep-seated iron-bearing magma. The molecular replacement of the schist and of breccias by ironstone showing the original structure, is evidence of an active circulation of water and of considerable chemical activity, and is such as is likely to have resulted from deep-seated solutions. Probably silica has been removed from the zone of ore deposition and redeposited, along with a proportion of ironstone, to form the cherts, but at a later stage these chert bodies have been fractured with the introduction of iron into the cracks. Simultaneous origin is shown by the merging of hæmatite into ferruginous chert.

If descending solutions from material formerly overhead be considered, there would be the objection that such large bodies as the Iron Monarch would call for a very much larger bulk of the ferruginous chert to furnish the iron than appears likely to have existed, and under atmospheric conditions this ferruginous chert is apparently very stable.

In the case of the schists the oxidised material, when compared with the few examples of unweathered schist exposed at Iron Monarch, appears rather to have been iron-enriched than impoverished.

On the whole it appears most reasonable to assume that these bodies have their origin in deep-seated solutions, and consequently may be expected to have a considerable vertical range.

The intensity of fracturing and the physical characteristics of the rock at the time of fracturing, have had a pronounced influence on the ore bodies. In the case of Middleback, the fracturing has been along definite lines, and the intervening rock has been insufficiently shattered to permit the free circulation of the mineralising solution, and so replacement has been incomplete.

At the Iron Monarch shattering has been much more intense, but there are certain well-defined zones, striking north-west and south-east and dipping north-east, of a somewhat unctuous clay, which represent beds of argillaceous rock that did not permit the formation of fractures in them, or if such fractures formed, they closed prior to mineralization, and so these beds were not permeated and replaced by the iron-bearing solutions to the same extent as the mass of the hill.

Through the channels formed by this shattering, iron-bearing solutions—probably alkaline carbonates—have made their way, dissolving the rock adjacent to the fractures and replacing the schist molecularly, so that even the highest grade ore may show the structure of the original schist. In places the replacement is incomplete, shown either as incomplete replacement of schist, or as a network of iron ore enclosing the untouched schist.

Kaolinization of the argillaceous schist appears to have been a widespread feature, though it is possible that the alteration is due to simple weathering.

The composition of the ore is indicated in the list of analyses on page 53 which are of chips collected at regular intervals from outcrops or from faces exposed in road cuttings. An exception is the sample of ochre (No. 3), which is a grab sample from a pile.

The samples, except those at depth, show the presence of carbonate of lime, which forms a coating or skin over much of the superficial ore. This travertinous limestone does not persist in depth, and probably has its origin in the withdrawal by capillarity of a little carbonate from the soil to be deposited on the evaporation of the water, by which it is carried in solution.

The superficial travertine is so widespread in South Australia that it has no special significance, except that it slightly reduces the grade of the samples; but the analyses show that the amount was much less than appeared probable from inspection.

The ore, however, was noted to shade into chert at the ends of shoots, and, furthermore, contains inclusions of unreplaced rock, some of which will lower the grade slightly, but some of the inclusions are large enough to be handled as waste.

The ore was observed on the surface over a vertical range of 400ft., and this, with the additional evidence of the shaft, indicates that the ore may be safely counted on to persist to the depths indicated on the sections on pages 55 and 56.

As lenses of ore are followed out to the end the ore was observed to become more siliceous. Transversely the bodies appeared to be clearly separated from the country in most cases where exposures were visible, but in some places replacement following the joints and cleavages had been incomplete, hæmatite enclosing masses of schist.

Middleback Range.

Location of samples in table on page 53.

- No. 1. Ore body in west crosscut, 108ft. level, on M.L. 2387.
- No. 2. Outcrop west of shaft on M.L. 2387.
- No. 3. Pocket of red ochre at 63ft. V.D. in shaft on M.L. 2387.
- No. 4. Hard blue hæmatite over 45ft. width of central body on road on M.L. 2398.
- No. 5. Red and black hæmatite and limonite over 1 chain on road, 4 chains south-west of east boundary of M.L. 2399.
- No. 6. Blue hæmatite over 1½ chains on road, M.L. 2399.
- No. 7. Medium-grade hæmatite on road for 3 chains in M.L. 2399 and half a chain in M.L. 2401.
- No. 8. From 9 chains along road crossing north-western ore body of M.L. 2401.
- No. 9. From 11 chains along road crossing main southern ore body of M.L. 2401.
- No. 12. Limonite over schist between west boundary of M.L. 2402 and peak on M.L. 2402.
- No. 13. Fairly massive hæmatite from ore body forming peak on M.L. 2402.
- No. 14. Hæmatite (schistose) from eastern ore body on M.L. 2402.
- No. 16. Floaters for 7 chains west of east boundary of M.L. 2400 on east-west ore body of spur.
- No. 17. Soft blue hæmatite of western ore body on M.L. 2392.
- No. 19. Hard hæmatite, eastern ore body at peak on M.L. 2392.
- No. 20. Blue schistose ore of eastern ore body on M.L. 2389.

No.—	1	2	3	4	5	6	7	8
Description.	Hæma- tite.	Hæma- tite.	Red Ochreous Hæma- tite.	Hæma- tite.	Hæma- tite.	Hæma- tite.	Hæma- tite.	Hæma- tite.
HO at 100° C. . .	% ·17	% ·10	% ·17	% ·05	% ·36	% ·08	% ·39	% ·18
H ₂ O at over 100° C.	·81	·41	·62	·91	2·42	·98	2·38	1·51
SiO ₂	1·68	1·10	4·02	1·48	1·10	·90	5·26	1·60
Fe ₂ O ₃	90·97	96·14	89·11	90·15	91·64	94·11	84·56	91·78
FeO	1·56	·30	Nil	1·35	·49	·52	·43	·49
Al ₂ O ₃	3·86	1·47	5·15	1·27	3·48	2·10	5·65	4·23
CaO	·44	1·00	·88	2·00	·80	1·36	1·00	·92
P ₂ O ₅	·08	·06	·02	·05	·11	·06	·06	·04
S	Nil	·04	·06	·08	·04	·04	·11	·06
TiO ₂	·01	Nil	·01	·02	Nil	·02	·23	Nil
MnO ₂	·84	·04	·09	·44	·04	·03	·03	·07
CO ₂	Nil	Nil	Nil	1·69	·11	·24	·37	·17
Total	100·42	100·66	100·13	100·09	100·59	100·44	100·47	100·05
Total Fe	64·89	67·53	62·38	64·17	64·53	66·27	59·52	64·62
Total P	·03	·02	·01	·02	·05	·02	·02	·02

No.—	9	12	13	14	16	17	19	20
Description.	Hæma- tite.	Limo- nite.	Hæma- tite.	Hæma- tite.	Hæma- tite.	Hæma- tite.	Hæma- tite.	Hæma- tite.
H ₂ O at 100° C.	·06	·22	·34	·26	·11	·14	·25	·12
H ₂ O at over 100° C.	·46	10·88	1·33	·54	1·18	·65	1·15	·32
SiO ₂	2·40	6·00	2·32	2·60	1·08	·70	1·34	1·02
Fe ₂ O ₃	93·00	73·30	93·50	91·61	93·72	95·69	94·50	94·42
FeO	·90	·24	·60	·98	·40	·29	·33	·19
Al ₂ O ₃	2·95	5·90	2·15	2·41	2·89	·40	2·10	2·89
CaO	·76	·52	·36	1·00	·92	·56	·36	·92
P ₂ O ₅	·02	·07	·11	·05	·04	·07	·19	·01
S	Nil	·10	·05	·02	·06	·08	·17	·07
TiO ₂	Nil	3·12	Nil	Nil	Nil	Nil	Nil	Nil
MnO ₂	·24	·02	·06	·47	·05	·05	·05	·03
CO ₂	·14	·14	Nil	·64	·21	·21	·08	·60
Total	100·93	100·51	100·52	99·98	100·66	98·84	100·52	100·59
Total Fe	65·80	51·49	65·70	64·88	65·91	67·50	66·41	66·23 Less than
Total P	·01	·03	·05	·02	·02	·03	·08	·01

NOTE.—The ferrous oxide figures include the calculated correction for the oxidizing action of the manganese dioxide present.

W. S. CHAPMAN, Analyst and Assayer.

The most notable feature of the analyses is their uniformity in iron and their purity and freedom from deleterious material. The average value of all analyses, with the exception of special samples Nos. 3 and 12, is:—

	Per cent.
Metallic iron	65·72
Phosphorus	·03
Metallic manganese	·11

Sulphur is present in small amount, and is probably derived from the small quantity of gypsum found in the soil.

No. 12, of the limonite capping, shows the limonite to be high grade, and to contain an abnormally high proportion of titanium. Owing to the relatively small proportion of limonite, the proportion of titanium will be negligible in the ore as mined.

The ore is fairly dense, but not as hard as that of the Iron Monarch, and may show evidence of the structure of the schist which it has replaced. The structure is generally finely granular, but there is a bluish variety which shows a slightly micaceous habit.

A small body or seam of high-grade red ochre of great staining power was cut in the shaft at a depth of 63ft. Sample 3 was taken from a heap of this material. There is also a lower-grade yellow ochre exposed on the south boundary of the Iron Baron. In exploiting the deposits it is probable that bodies of these ochres may be disclosed, bodies that will have a greater value as an ochre than as an ore of iron.

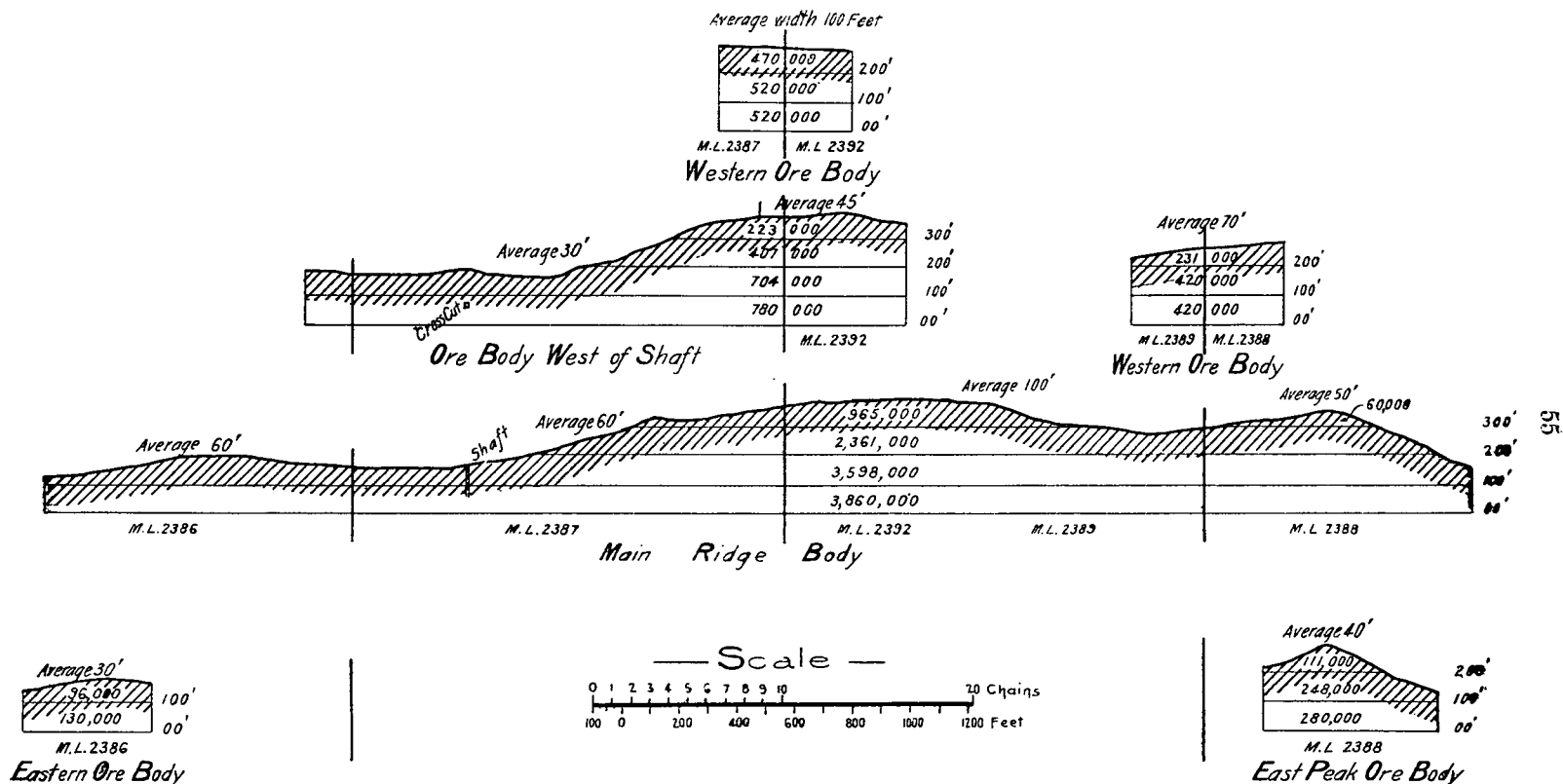
The ore being considerably harder than the decomposed schist in which it occurs, forms, in the Iron Prince group, the crest of the ridge, peaks, and spurs. In the Iron Baron the presence of two important lines of ore bodies some distance apart has resulted in the formation of a plateau between them, for the most part covered by a lateritic or limonitic crust 1ft. to 3ft. thick, derived from the underlying ferruginous schist. Wherever the bottom of this crust was observed it rested on schist. The hæmatite is not limonite encrusted, but persists to surface. In places it forms bold crags and scarps, but usually the outcrops are but little higher than the enclosing rock, and the line of demarcation is very obscure, owing to the large quantity of hæmatite strewn about the surface.

. For this reason the presence of the smallest particles of slate in the debris was taken as evidence that the limit of an ore body had been reached, and the outcrops have been drawn on this assumption. On the other hand, the road cuttings show in the ore bodies inclusions of schist, of which little or no evidence is visible on the surface, and in some cases the widths of outcrops shown may be under-estimated in consequence, as may be their extension from the position where crossed by a traverse. For these reasons, after estimating the area of outcrops and the tonnage, the writer considers a range of ± 25 per cent. in the tonnage justified in the present state of exposure of the ore bodies.

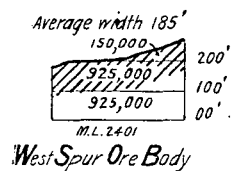
Only ore bodies of an estimated average width of 30ft. or over were taken into account, as lesser bodies would hardly be worked independently. Some of the lesser bodies will, however, contribute some ore, as they will be laid bare in the battering of open cuts on the larger bodies.

The total extent of these larger outcrops is 31 acres, extending over an aggregate length of 17,180ft. by a mean width of 78ft., and their ore bodies should contain a total of 14,890,000 tons within 100ft. of the surface.

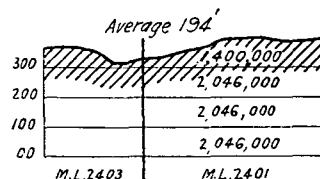
The deposition of the ore has been influenced by the fracturing, and replacement has taken place along and close to the fracture. In some cases, notably in a bare patch near the boundary of M.L. 2399 and 2401, cross-fractures filled with ore form a network in the schist, which has been in part replaced. In other portions (north end of M.L. 2387) a breccia identical in structure with the brecciated schist zones of M.L. 2388 has been completely replaced by hæmatite, preserving not only the form of the breccia, but the foliæ of the component schist. In M.L. 2402 the eastern body shows contortion of the iron ore structurally identical with the foliæ of the schist, and there is no doubt but that the ore bodies are due to replacement of the schist by hæmatite.



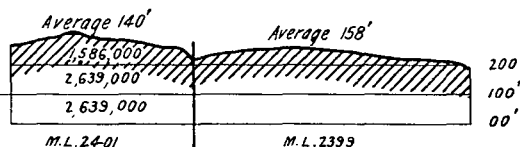
Estimated Gross Tonnages and Widths of Ironstone in Iron Prince Leases.



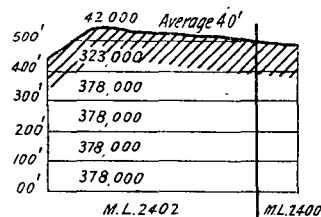
West Spur Ore Body



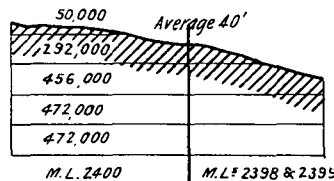
South Western Ore Body



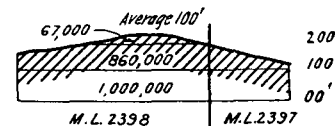
North Western Ore Body



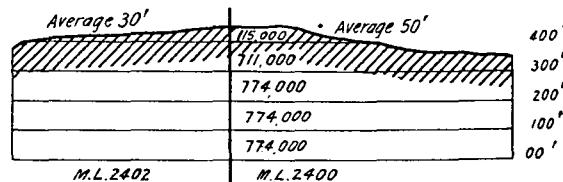
Ridge Central Ore Body



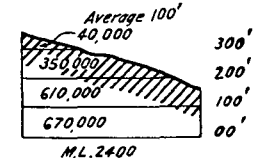
Ridge Ore Body



Northern Ore Body



Eastern Ore Body



Eastern Spur Ore Body



Estimated Gross Tonnages and Widths of Ironstone in Iron Baron Leases.

From the mode of origin there is very little doubt but that the ore bodies persist at least as far as the level of the plain, and probably to greater depth.

The deepest ore seen is that of one of the lesser ore bodies which is cut in the west crosscut of the shaft at about 60ft. above datum level. Natural exposures are not seen within 100ft. of datum, but this is because the ends of the deposits do not persist north or south into low ground, and so direct evidence could not be got.

The attitude of the ore bodies is of importance in future work. Most of the ore bodies, especially those on the western side, are within 10° to 20° of vertical, with a slight tendency to underlie to the westward. The south-eastern bodies are also apparently vertical, but the foliæ of the replaced schist shows a distinct westerly dip. This, however, is probably within the ore bodies, which themselves appear to be nearly vertical. The east-west spurs are vertical.

Tonnage.

The blocks on pages 55 and 56 show the longitudinal profile of the various ore bodies of the Iron Prince and Iron Baron.

Tonnages have been worked out on the basis of 9 cubic feet to the ton, and on the assumption that the ore bodies, on the average, maintain their average size in depth, for:—

- (a) Within 100ft. of the surface.
- (b) For each 100ft. above datum level.

The former shows a total of 14,890,000 tons of ore. For reasons set forth above, a factor of safety of ± 25 per cent. has been adopted for this giving a possible minimum of 11,170,000 tons and a possible maximum of 18,610,000 tons. The road cuttings show that there are inclusions of country or waste within the bodies, but these may be offset by ore in small bodies not taken into account, and which will be won during the removal of wall-rock to a safe angle for open-cut work. In taking a depth of 100ft. it is considered that this will be the approximate limit of open-cut working. Some of the narrower bodies will certainly not be worked open-cut to this depth; on the other hand, at least three bodies could be carried to considerably greater depths.

Below this depth there is undoubtedly a considerable tonnage of iron-stone, and the gross quantities were worked out for each 100ft. above datum level. The recovery of such ore implies the use of underground methods, and as individual bodies may pinch out without other compensating bodies being developed, and as poor zones and pillars would be left, the writer considers that from the gross probable ore indicated on the plan (43,170,000 tons) there should be deducted 50 per cent. of that in the zone 0-100ft. above datum, and 25 per cent. of that in the zone 100ft. to 200ft. above datum, giving a net quantity of 32,151,000 tons recoverable above datum level, by both open-cut and underground mining.

In the present state of knowledge of the deposits a variation on this net tonnage of ± 25 per cent. is considered possible, giving the probable recoverable ore above datum as somewhere between the limits of twenty-four and forty million tons.

It is recognised that, with the large tonnage available for open-cut mining on this property and in the Iron Knob and Monarch ore bodies, amounting to 148 million tons, and with an output of about a million tons of ore a year, the period of underground mining is far in the future. Yet, in view of the gradual reduction in grade of iron ores used throughout the

world, it is inevitable that the ore inaccessible by open-cut workings will be mined by underground methods.

The ultimate utilization of this ore should not be lost sight of, so that due provision may be made for the disposal of waste from open-cut workings in such localities as not to hamper future operations. The sites available for waste dumps are evident from examination of the plan of the outcrops, and are for the most part on the eastern slopes.

The waste will not come to any extent from overburden proper, the bodies persisting to surface or to surface soil, but will be obtained in cutting back the slopes of the open cuts to safe angles. During such cutting back a proportion of ore from minor bodies not taken into account will be recovered.

The Ores.

The predominant ore is hæmatite, but a crust of limonite, of which the maximum observed thickness was 3ft., has formed above schist over several acres.

Conditions Affecting Mining.

The chief disability is the lack of water, which will have to be supplied before serious exploitation of the deposit is possible. All the evidence is against the prospects of useful wells, and dependence will have to be placed on surface supplies. The average annual rainfall is between 9in. and 9½in., of which about two-thirds falls in the six winter months.

There are catchments to the eastward that would supply excavated dams, while the creek at Old Middleback Station should provide an efficient if small catchment provided a suitable reservoir site can be located.

MAGNETITE NEAR BENOWRIE OR BOOLCOOMATTA STATION.

Locality, 3½ miles west of Benowrie Station, half a mile south of the Benowrie-White Well road, and 19 miles north-west by west of Mingary.

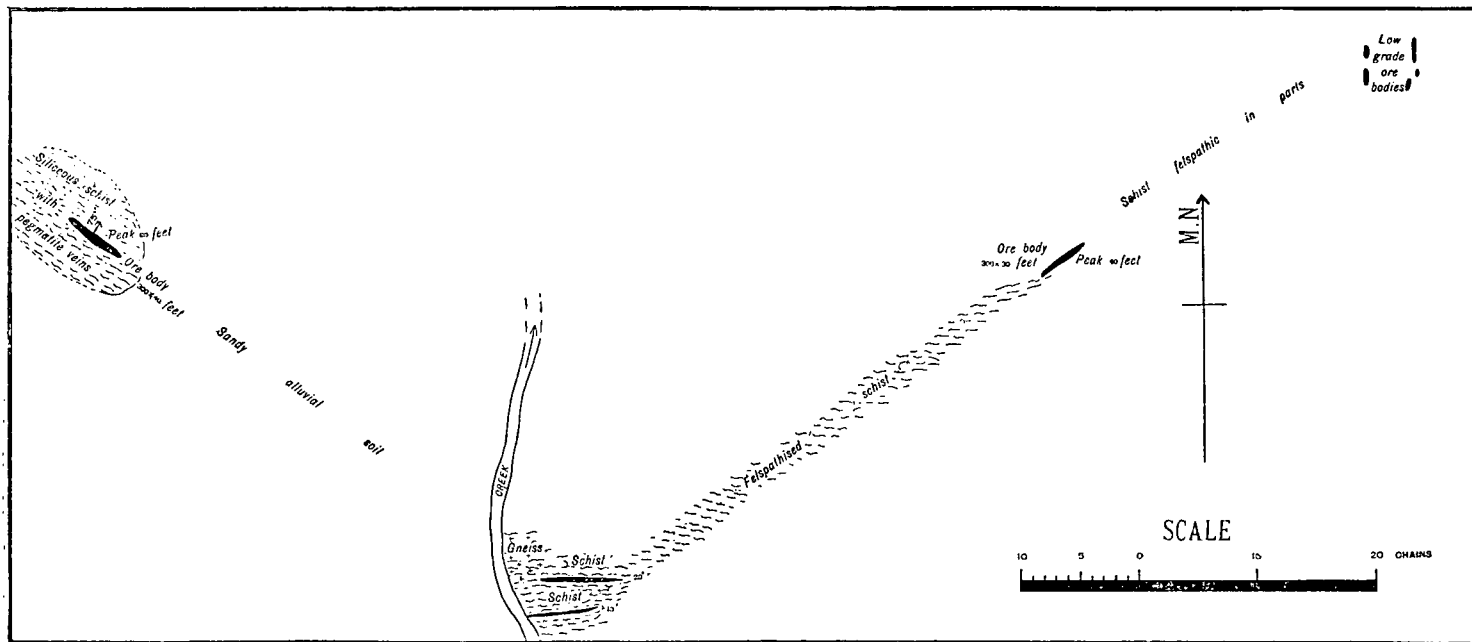
A small lode of hæmatite and magnetite was formerly worked for flux at this point, the material being carted to Mingary for use at Broken Hill.

The country consists of slate, phyllite, and more or less completely feldspathised schist striking approximately north 83° east, and dipping at high angles. The lode, lenticular in habit, strikes N. 70° E., and dips steeply to the south-south-east. It has a length of 170ft. and a width of 8ft. About half is good grade, the balance becoming more and more siliceous towards the ends. Massive quartz is present 100ft. east-north-east of the ironstone, and lower-grade ironstone is visible about 700 yards farther on.

A shaft about 50ft. in depth has been sunk through banded silicified slate to the lode, and some stoping is said to have been done. The ironstone is reputed to have carried 1oz. to 2oz. of silver per ton, but none was present in the general sample taken by the writer. A general sample from the outcrop and shaft dump yielded:—

	Per cent.
Water	.40
Insoluble	1.63
Iron	69.13 = 98.76 of Fe ₂ O ₃
Alumina	2.08
Phosphoric anhydride	.29
Titanic oxide	.23
Lime	.20

About 150 yards to the south-west of this shaft, shallow workings, stated to have been made for bismuth, expose amorphous graphitic slate. A ton of bismuth is stated to have been won from these workings about 30 years ago.



Plan showing relation of Ore Bodies at Billeroo.

BILLEROO IRONSTONE DEPOSITS.

Locality, 5½ miles north-east of Mount Victoria trig., and 34 miles north-west of Olary railway station.

The Billeroo deposits are enclosed in schists, felspathised in varying degree, and traversed by pegmatite. The rocks are probably of Pre-Cambrian age. The ironstone consists of magnetite and hæmatite, and the deposits show evidence, by the presence of residual laminae and zones of undigested siliceous material, of having been formed by replacement of the more soluble constituents of the country rock by ironstone. The bodies may, therefore, be expected to have considerable vertical extension. In consequence of the incompleteness of the replacement the quality is not very good.

Four deposits occur, and their relative positions are indicated on the accompanying sketch plan.

The western deposit is the largest, and forms the backbone of a hill rising about 80ft. above the surrounding plain. Lenticular in plan, the body has a length of 360ft. and a width of 40ft., and forms a bold outcrop. It is enclosed by siliceous schist and schist dipping with the deposit 75° to N. 30° E. The outcrop proper contains 30,000 tons to 40,000 tons of ore, and the body may be taken as, at 10 cubic feet to the ton, containing 1,440 tons per foot of depth, and a total of 170,000 tons to a depth of 100ft. below the surface, assuming that the body maintains its dimensions to that depth.

An average sample broken from this outcrop yielded:—

	Per cent.	
Water	2.42	
Insoluble	26.04	
Iron oxide	71.15	= 49.81 % of iron
Alumina	Trace	
Lime	.14	
Phosphoric anhydride	1.11	
Titanium dioxide	.09	

Phosphate of lime was noted on the flatter surfaces of this and other outcrops of the region, and has its origin in bird guano. It ranged in thickness from a greyish-white deposit ¼ in. thick to a glossy film, and while of itself negligible, may, in any assays of outcrop samples, cause an unreliable estimate of the phosphoric content to be formed. In sampling, any apparent deposit of this material was excluded.

A little over half a mile south-east by east from this deposit, and separated from it by alluvium, two bodies of ironstone are present in felspathised schist, which here strikes east and west.

The northern body has a length of 7 chains and a width of about 20ft., and does not form a bold outcrop. The grade is about the same as the first described body, and the tonnage to a depth of 100ft. may be estimated at 90,000 tons. Three chains south of this the second body forms a wall-like mass 10ft. thick and about 100 yards in length. The grade is less than that of No. 1 by perhaps 10 per cent., or say 40 per cent. of metallic iron, and the tonnage above 100ft. may be taken as 20,000 tons to 30,000 tons.

The third deposit is 50 chains north-east by east from the last, and is about 300ft. by 30ft., culminating in a peak 40ft. above the surface. It is the best grade deposit, and was estimated to be 10 per cent. better than the sample of the first deposit, or, say, 60 per cent. of iron. The tonnage is approximately 10,000 tons in the outcrop and 90,000 tons to a depth of 100ft.

The fourth or eastern deposit consists of a group of five small outcrops, distant 34 chains north-east by east. The individual bodies are small and apparently disconnected, and too low grade to consider as ore.

The total probable tonnage of the Billeroo deposits above 100ft. in depth is as follows:—

Western deposit	170,000
Southern deposit, north body	90,000
Southern deposit, south body	25,000
Third deposit	100,000
Total	385,000

The grade is approximately that of the first deposit, or about 50 per cent. of metallic iron.

GRANT'S AND CUTANA QUARRIES.

NEAR CUTANA, ON THE BROKEN HILL RAILWAY, 194 MILES
FROM PORT PIRIE.

A large area in this locality was at one time held by the Koolka and Mingary United Ironstone Flux Company, Limited, which built a tramway from near Cutana about 1892 to transport flux to the smelters at Broken Hill.

The leases were abandoned when the Iron Knob ore became available, and the rails were removed.

The deposits are scattered, and consist of brown iron ore and a little hæmatite. The workings were numerous, and extended over a considerable area, as may be seen from the accompanying sketch plan.

The metalliferous area is slightly undulating, and is occupied by sediments, considerably metamorphosed and in places fclspathised. For the most part the sedimentary beds dip at low angles, and the ironstone is conformable with them in strike and dip. Marly rubble is abundant as a superficial covering, and the ore may have resulted from the solution of ferruginous limestone. No workings have been carried deep enough to determine this point.

The ore bodies are definitely enclosed in the country rock, and are not subsurface laterites resting unconformably upon the edges of the primary rocks. The ironstone deposits are found in the more elevated portions of the area, but this is due to the superior resistance to erosion of the outcrops, and does not indicate that they are residual patches of a larger and formerly continuous sheet. Whether the several outcrops represent individual and superimposed beds, or whether there has been a somewhat complex system of step faulting which has resulted in the repetition of the same bed on the surface, was not established, and in the absence of outcrops probably can only be solved by boring.

The ridges are strewn with stones and nodules of brown iron ore, and where pits have been sunk through the clay and marl, bodies of ironstone are cut beneath these surface indications.

Description of the Workings.

The workings are very scattered and occupy a number of zones in which quarries that produced flux and many subsidiary prospecting pits have been sunk. Many of the latter were stopped as soon as the iron ore was reached, and the quality recorded is probably, judging from the behaviour of the ore bodies in the deep workings, below the true average of the

underlying body. For the same reason the thickness was exposed in only a small proportion of the openings, so that estimates of tonnage are difficult to make with any degree of accuracy.

The south-easternmost exposure forms a low rise trending east-north-east, on which are several small pits, exposing low to medium grade limonite beneath marl and ferruginous clay. Eight chains to the east-north-east, across a soil-filled depression, is another rise on which good grade limonite is exposed in two small pits. These two exposures appear to be on the same zone, which has, according to shoad exposure, a width of about 2 chains, and a length, including the soil-filled depression, of 16 chains.

The next deposit to the northward has been proved by prospecting pits over an area 18 chains long and 2 chains wide. The general trend is north-east. On this body, *at the terminal point of the tramway*, is one of the more important quarries. Over about one square chain the ore has been removed to a depth of 12ft., equivalent to an output of about 4,000 tons. The ore bed dips 20° to the west-north-west. The ore is somewhat concretionary, and shows a rude stratification and small interbedded lenses of sandstone.

The following section is visible in the quarry:—

	Ft.	Ft.
Marly rubble	0 to 1	
Red clay	1 to 7	
Red clay and rubbly limonite	7 to 10	
Ironstone with thin lenses of sandstone	10 to 22	
Red, yellow, and brown ochreous clays	22 to floor	

A shaft, situated a chain west of this open cut, and open to 20ft., exposes 15ft. of clay and 5ft. of iron ore.

Two samples were taken from the open cut and one from ore stacked on the surface.

	Lower Six Feet of Ironstone Bed In Quarry.	Upper Six Feet of Ironstone Bed in Quarry.	Heap on Sur'ace.
	Limonite and Hæmatite.	Limonite.	Limonite.
	%	%	%
Water	7.84	9.18	8.62
Insoluble	21.16	24.81	24.83
Iron	46.61	42.79	43.46
Equivalent to ferric oxide	66.59	61.13	62.09
Alumina	3.34	3.49	3.25
Phosphoric anhydride	0.03	0.04	0.04
Titanium dioxide	1.02	0.90	0.92
Lime	0.22	0.16	0.22

Eight chains to the eastward of this deposit limonite of medium grade is exposed by test pits and an open cut 12 ft. deep.

These workings prove an area of over an acre, but the boundaries of the deposit do not appear to have been reached. It cannot be stated that this deposit is identical with that on which the quarry sampled above is situated, as the intervening hollow is soil-filled, and the deposits are not co-linear.

A small pit 14 chains to the north-north-west of the "Terminus Quarry" exposes limonite of fair grade, but the next workings of importance are situated $23\frac{1}{2}$ chains from the above-mentioned quarry.

This group of workings exposes iron ore for a distance of 14 chains in an east-north-east—east-south-east direction, over a width of $1\frac{1}{2}$ chains to 2 chains. The body dips 20° N.N.W.

The principal excavation is a little over 200ft. in length, 40ft. in width, and 12ft. to 15ft. in depth. From 10ft. to 14ft. of iron ore is exposed, and an average sample taken from the walls showed this ore to have the following composition:—

	Per cent.
Water	5.32
Insoluble	17.84
Iron	51.34
Equivalent to ferric oxide	73.34
Lime	.26
Phosphoric anhydride	.05
Titanium dioxide	1.66

A spur from the tramway extended into this quarry, and good grade ore is exposed in the cutting for a distance of 130ft. The other openings on this body of iron ore are shallow prospecting pits, and the ore exposed in them is low to medium grade. The quality probably improves in depth as these pits do not do more than penetrate the marly superficial overburden.

After following the tramway formation for 35 chains in a west-north-west direction, passing an outcrop of quartzite dipping 45° N.N.E., a group of small workings is reached. These workings expose ironstone in an area 2 chains to 3 chains wide, and 10 chains long, the longer dimension trending in a north-north-west direction. The ironstone is partly hydrated and partly consists of hæmatite, and varies considerably in quality, portions being somewhat siliceous. The deposit is made up of bodies of ironstone separated by decomposed argillaceous rock, and dips 35° to 45° to the north-east. Some hundreds of tons of ore have been removed, involving the shifting of a very considerable proportion of waste.

Five chains to the eastward of this group of workings a cutting exposes veins of iron ore 6in. to 12in. thick, separated from one another by beds of clay of equal or greater width, the whole body dipping 60° N.E. The ridge on which these workings have been made trends to the north-north-east and east, and several shallow prospecting pits expose low-grade limonite over a distance of 18 chains. The exposures are insufficient to determine whether there is a large tonnage available.

At 11 chains in a north-westerly direction along the tram formation from a subdivisional fence, small prospecting pits expose low-grade limonite $3\frac{1}{2}$ chains to the south-west, and at 29 chains from the fence and $6\frac{1}{2}$ chains south-west a pit 20ft. by 40ft. exposes fair-grade limonite dipping 40° N.E.

Twelve chains north-west of this opening a group of quarries extends 9 chains to the south-west. Good to medium grade limonite, interbedded with clay, dips 60° to N. 30° E. in the easternmost of these workings. In the others the attitude of the limonite is obscure. A fair tonnage of flux appears to have been won here, and, as at the other workings, a considerably greater tonnage remains untouched.

This body, 40 chains N.W. of the intersection of the fence and tramway formation, is not shown on the plan on page 62.

Beyond this group the workings are of less importance. A small amount of work has been done on the north of the tram line at 48 chains N. 33° W.

A mile W. 15° N. from this point a costean exposes low-grade hæmatite and limonite exceeding 30ft. in width at the north-eastern end of an ironstone-strewn rise extending for 11 chains in length and for a width of about 1½ chains. An outcrop of hæmatite 100ft. in length and 20ft. wide occurs in this area, but a small costean and a pit show that the hæmatite is associated with schist and would require to be separated.

Three-quarters of a mile to the westward are the group of workings known as the Cutana quarries. One pit about 66ft. by 20ft. by 10ft. shows decomposed schist dipping about 40° to the north-east, and containing interlaminated limonite as well as limonite-filled transverse seams. A little over half an acre round this pit is evidently similar in grade.

The only other working of any magnitude is 8½ chains to the north, and consists of a pit 100ft. by 20ft. x 10ft., exposing schist and arkose quartzite, with irregular pockets and veins of limonite. The proportion of waste is large.

A feature of these workings is the large amount of crystalline gypsum in the superficial soil and the associated schist.

Several minor openings expose similar material, but the grade is very low. The tram line was not built to these deposits, though they are nearer the railway than those that were worked.

Tonnage and Grade.

Any attempt to calculate the tonnage and grade cannot be much more than an estimate owing to—

- (1) The insufficiency of exposures;
- (2) the variable proportion of waste with the ore;
- (3) probability that the limonite is found near the surface only, and may give way to leaner rock at shallow depths.

In assuming the dimension of depth of iron ore a figure is taken for the ore itself. The quarry to obtain this may be double the depth assumed for the ore, as an arbitrary limit of a volume of waste equal to the iron ore has been taken as possible in working.

The grade is estimated for several quarries, being based on the difference in appearance of the ore from that of the portions sampled.

Ore Body.	Acreage.	Depth of Recoverable Ore Assumed.	Estimated Grade.	Tonnage.
		Feet.	%	
South-easternmost	3.2	10	40	140,000
Terminus	3.6	15	43	235,000
East of terminus	1.0	12	40	50,000
North main quarry	2.8	12	45	126,000
On subdivision fence	2.0	10	50	87,000
40chs. N.W. of subdivision fence	1.5	15	55	107,000
Hæmatite and limonite outcrop (<i>see</i> p. 65, second paragraph)	1.1	15	45	65,000
	15.2	—	45	810,000

To this the smaller bodies might contribute to a slight extent, but the exposures show a considerable proportion of waste, and it appears improbable that there would ever be much work done outside the seven areas enumerated above.

There is little doubt but that boring to shallow depths would prove a very considerable additional tonnage, and also that the workable distances down the dip might, in some cases, be increased.

The deposits must, however, be regarded as discontinuous beds, each of which would require to be attacked separately, and there is no justification for assuming that the scattered workings prove a continuous deposit. It may have been at one time, and been repeated on the surface by faulting, or the beds now outcropping may be superimposed. In any case the limit to working is the extent of overburden, and whether the bodies are faulted portions of one bed or the outcrops of several, is of no great practical moment at the present day.

MINERAL SURVEYS 19388 AND 19389.

LOCALITY 10 $\frac{1}{4}$ MILES SOUTH OF MINGARY RAILWAY STATION.

This deposit was formerly worked for flux for the Broken Hill smelters, and appears to have produced between 5,000 and 8,000 tons of flux, which was carted to Mingary. It is now abandoned.

A quarry, approximately 100ft. square, has exposed a bed of turgite dipping in a north-westerly direction, and resting conformably upon a somewhat knotted schist dipping 30° to the north-westward. This schist in turn rests on mica schist. The limonite bed has a flatter dip on the north-west side, and is exposed by the quarry and a cutting on the north-west side for a width of 185ft. The workable ore is covered by 2ft. to 6ft. of rubbly marl, including a considerable proportion of lime-coated limonite. The maximum thickness exposed is 18ft., and the quarry shows an average of 15ft. Bands of clay up to 2ft. in thickness are interbedded with the limonite, the whole body showing a rude stratification. The ore itself is pisolitic in structure, and contains a large proportion of voids.

A typical sample of the material as sorted for shipment yielded:—

Water.....	3.66
Insoluble matter	11.32
Iron	57.09
Equivalent to Fe ₂ O ₃	81.56
Alumina.....	2.06
Phosphoric anhydride	0.10
Titanium dioxide	1.66
Lime.....	0.20

Secondary gypsum occurs in the clays, in joints, and in vughs, as seams up to half an inch in thickness.

The turgite is interbedded with the enclosing rocks, and possibly owes its origin to the weathering and solution of a ferruginous and calcareous bed, and cannot be expected to persist in depth beyond the zone of weathering. It does not form an outcrop, but superficial stones indicate that the body extends a little to the westward, and for 1,150ft. N. 60° E. to two small pits exposing good to medium grade limonite beneath marl, and for 300ft. beyond them.

A considerable tonnage evidently exists, but exposures are insufficient to enable any reliable estimate to be made. On the dimensions given by the quarry—a mean thickness of 15ft. and a width of 185ft.—if the deposit is of similar grade over the 1,500ft. along which indications are present on the surface, there should be approximately a quarter of a million tons, but this is little more than a guess, which would have to be confirmed by the sinking of trial pits if the body is seriously to be considered as a source of iron ore.

KOOLKA OR "IRONSTONE WELL" DEPOSITS.

ON MINERAL SECTIONS 19155 AND 19156.

Locality, near Ironstone Well, $7\frac{3}{4}$ miles north-north-west of Triangle Hill (Bimba) trig., and 29 miles north by west from Olary railway station.

The deposits consist of two main lenticular masses of magnetic iron ore about 35 chains apart, and some minor outcrops. The bodies are at right angles to one another and are not connected. They occur in schist, claystone, and sandstone superficially silicified in the vicinity of the ore bodies. As in the case of Billeroo, the deposits are due to metasomatic replacement, but the process has been more complete, and the ore is consequently high grade.

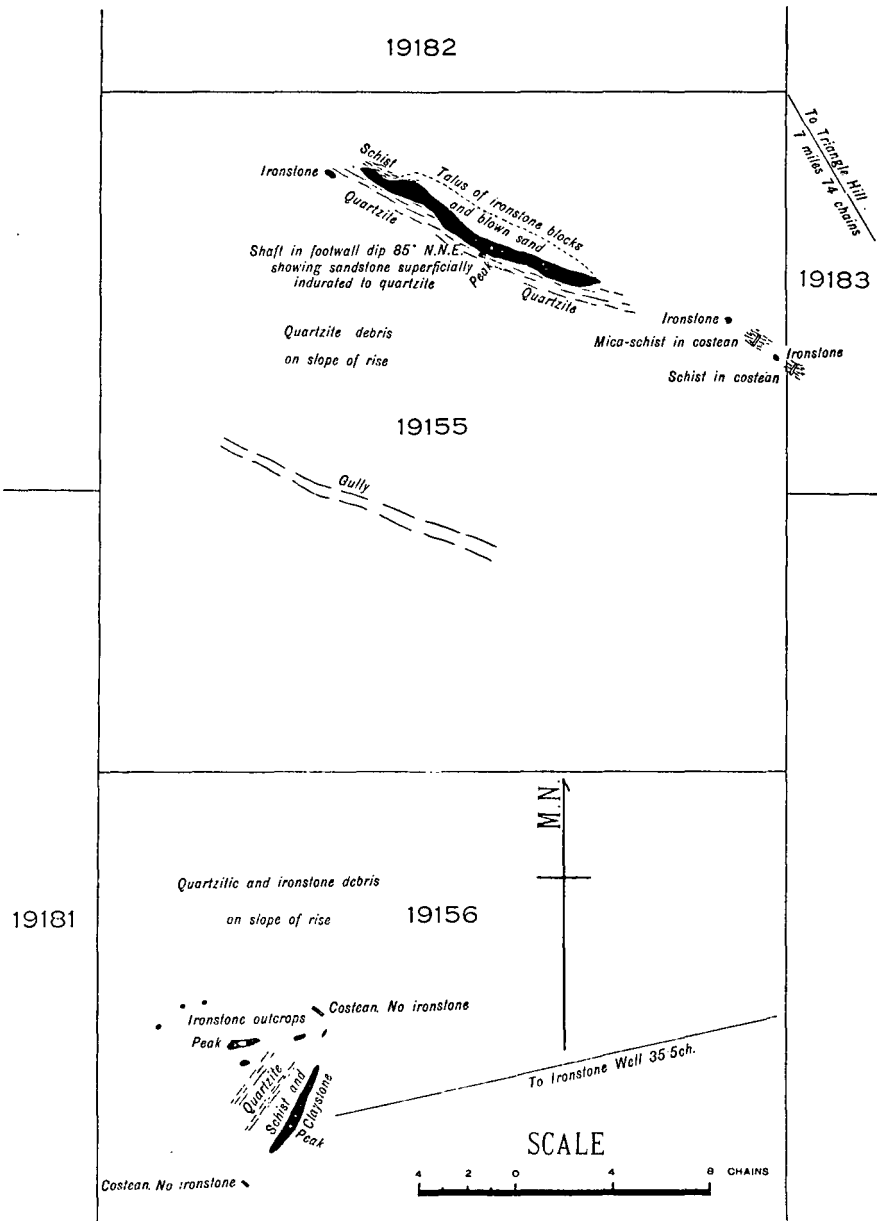
The main or northern deposit has a length of 11 chains, a mean width of 30ft., and culminates in a peak about 50ft. in height. It dips 85° to north-north-east, and is enclosed by schist on the northern side and by sandstone, superficially indurated to quartzite, on the south. The north side is flanked by a talus of ironstone blocks and wind-blown sand.

On the southern side a shaft has been sunk to a considerable depth (possibly 100ft.), following the footwall of the ironstone, and from the evidence of the dump a crosscut appears to have been driven into the deposit. The eastern 130ft. of the ore body is slightly lower grade, owing to the presence of bunches of vein quartz, but the proportion is unimportant. Costeans and the exposures at the ends of the mass show that there is no extension beyond the limits shown on the plan.

Tonnage may be estimated as 50,000 in the outcrop proper and 2,700 per foot depth, or a total of 320,000 tons above 100ft. in depth. From the type and mode of origin of the deposit there is no reason to suppose that it will not persist to a considerably greater depth, but underground mining would be prohibitive, and so a greater extension in depth cannot be taken into account.

Samples were taken from this body and from the southern deposit as follows:—

	Koolka North Body General Average.	Koolka North Body from Dump of Shaft.	Koolka South N.W. Peak Deposit.	Koolka South Deposit. South Peak General Average.
	%	%	%	%
Water	1.90	1.32	1.12	1.22
Insoluble	2.67	3.08	2.16	3.09
Ferric oxide	95.69	95.53	94.74	94.41
Equivalent to metallic iron..	66.98	66.87	66.32	66.09
Lime	.40	.24	.18	.08
Alumina	1.86	1.40	2.87	.64
Phosphoric anhydride	.45	.57	.55	.81
Titanium dioxide	.16	.10	.22	1.22



Plan showing Ore Bodies at Koolka.

The southern group consists of a lenticular body, striking north-north-east, 4 chains in length and 20ft. in width, and several smaller bodies. The largest of the smaller bodies is the north-west peak, which has a length of 90ft. and a width of 18ft. The lenses of ironstone are enclosed in claystone, schist, and quartzite, the latter being probably the silicified outcrops of sandstone beds, such as were exposed in the shaft at the north deposit.

Tonnage probable in the south deposit is as follows:—

North-west peak body	18,000
South peak body, 660 tons per foot of depth—	
Outcrop	10,000
To 100ft. in depth	66,000
	94,000

The analysis of samples from these masses are given in the table on page 67.

The other bodies of ironstone in the vicinity are so small as not to be worth taking into account.

The three main deposits at Koolka contain a total quantity of 414,000 tons above the 100ft. level, of an approximate content of 66.7 per cent. of metallic iron and .55 per cent. of phosphoric anhydride. The analysis and tonnage indicate that Koolka can only be utilised as a contributor to the supply of a basic open hearth steelmaking plant, or for pig iron. But under present conditions of transport it is not within economic radius of any existing or possible smelters, and the size of the deposit is such that the cost per ton of redemption of a railway built specially to serve it would be very heavy.

LEIGH CREEK COAL MEASURES.

NEAR COPLEY, GREAT NORTHERN RAILWAY.

In the area occupied by the Jurassic coal measures of Leigh Creek numerous limonitic outcrops occur interbedded with the carbonaceous shales and dipping at angles up to 30° from the horizontal.

Recent boring (No. 6 bore, 1919) has proved that these limonitic bodies are on the outcrops of beds of siderite, or carbonate of iron. The beds are thin, from an inch to 1ft. to 2ft., and consist of a hard, dark-grey impure carbonate of iron. An analysis (11092) by Mr. W. S. Chapman, of core from a bed at 199ft. to 200ft. in No. 6 bore showed the material to consist of—

	Per cent.	
Silica	13.60	
Alumina	5.92	
Ferric oxide	.24	} Metallic iron 36.67 %
Ferrous oxide	47.70	
Lime	.90	
Magnesia	.36	
Carbon dioxide	30.00	
Water, organic matter, &c.	Not determined	
	98.72	

This analysis and the mode of occurrence show that the material is very similar to the clay ironstones of Great Britain.

The migration of the limonite formed by the decomposition of this siderite and the impregnation of the adjacent beds, together with the low angle of dip, give the impression of larger bodies than they are proved to be when bored through.

IRON ORE NEAR WOOCALLA.

71 MILES N.N.W. FROM PORT AUGUSTA.

A body of iron ore is situated about three miles east of Woocalla and a quarter of a mile north-east of the Sweet Nell Copper Mine.

The southern end of the body shows in a saline area or lake bed, from which it can be traced up a low quartzite-strewn ridge on a bearing of N. 10° E. The deposit is said to be traceable for nearly four miles, and to be very similar to the southern portion that was seen by the writer.

Where seen the dip is about 25° to E. 10° S., and, as far as could be judged, the thickness of the bed does not exceed 10ft. The iron ore is probably interbedded with the Ordovician quartzite, as a small bed of similar ironstone occurs a little to the north-west, lower in the series, and has a U-shaped outcrop due to the erosion of a pitching anticline. The best portion of the main body seen had a length of 10 chains, and a sample chipped from the outcrop at intervals along this distance had the composition indicated in the first column. The second column is an analysis of a sample sent to the Department of Mines, from the same deposit, and analysed 19/12/1916.

	%	%
Silica	4.19	.92
Alumina.....	3.70	.98
Ferric oxide. Metallic iron 56.7%.....	81.0	90.83
Ferrous oxide	—	.26
Water (H ₂ O) at 100° C.	} 7.80	{ .05
Water (H ₂ O) over 100° C.		
Titanic dioxide	—	Nil
Phosphoric anhydride	.22	.09
Manganous oxide and undetermined (MnO ₂)	.09	1.53
Sulphur	2.05	.17
Barium oxide (BaO)	1.00	.17
	100.05	100.00

The ore consists of limonite, goethite (?), and turgite, and the silica present is in the form of small quartz grains. At either end of the portion sampled the iron content diminishes considerably, with a corresponding increase of silica, which occurs as quartz granules and quartzite pebbles. Northwards along the outcrop other shoots, similar to that sampled, are stated to be present at intervals in the lower grade material.

The outcrop disintegrates into black ironstone stones or "gibbers," and the occurrence of similar "gibbers," some of which are manganiferous, suggests that this or similar beds are present between the locality described and Woocalla.

Under present conditions the situation, composition, and relatively small size of the better-grade shoots do not warrant the exploitation of the deposit as an ore of iron.

OTHER LOCALITIES.

Iron ore is recorded as occurring at the following localities, but as far as could be learnt, in quantities insufficient to justify examination:—

County of Adelaide.—Barossa, Kersbrook, Para Wirra, and Talunga.

County of Burra.—Burra and Gum Creek.

County of Dalhousie.—Eurelia.

County of Fergusson.—Pine Point.

County of Flinders.—Port Lincoln.

County of Hanson.—East of Hawker.

County of Jervois.—Hundred of Hawker.

County of Light.—Kapunda.

County of Manchester.—Pandurra.

County of Victoria.—Crystal Brook, Gladstone, Huddlestone.

County of York.—Point Lowly.

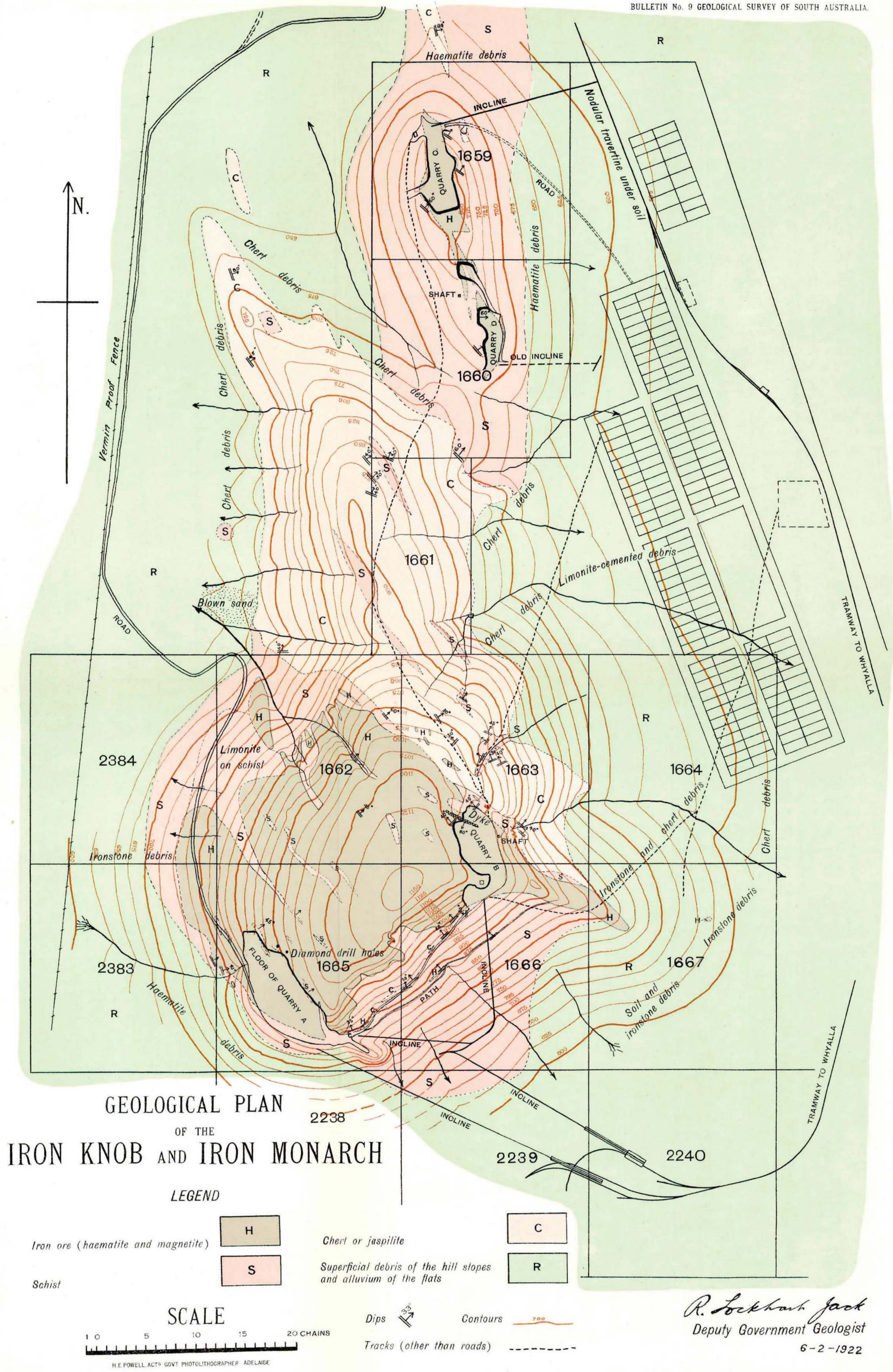
Outside of Counties.—Angipena, Beltana.

R. LOCKHART JACK,

Deputy Government Geologist.

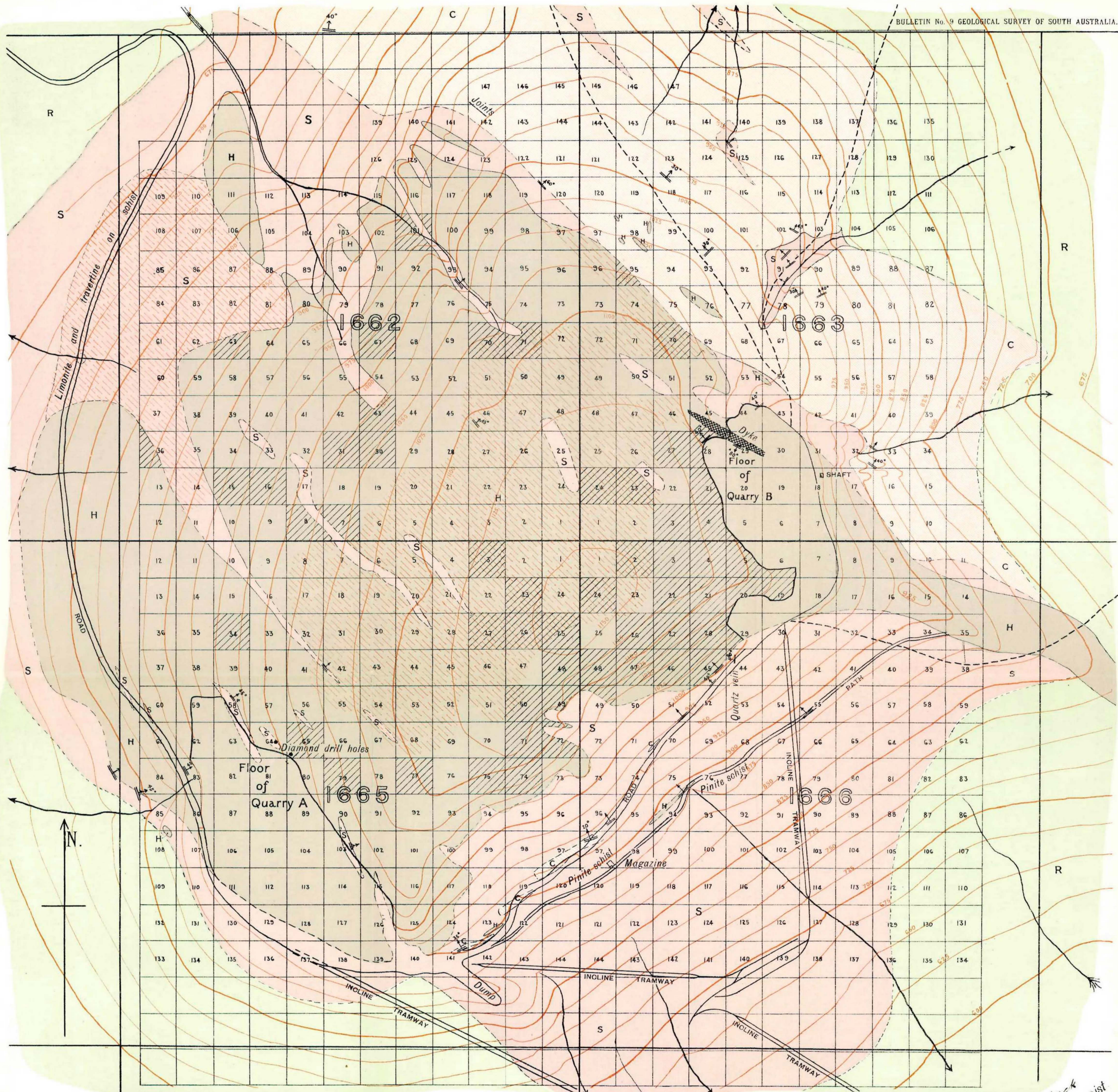
Adelaide, 6th February, 1922.





GEOLOGICAL PLAN
OF THE
IRON KNOB AND IRON MONARCH

R. Lockhart Jack
Deputy Government Geologist
6-2-1922



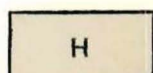
GEOLOGICAL PLAN OF THE IRON MONARCH

SCALE

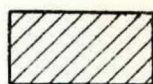


LEGEND

Iron ore (haematite and magnetite)



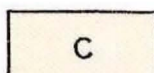
More than one per cent of manganese



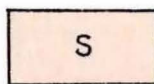
Dips



Chert or jaspilite



Schist



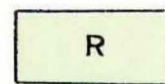
Contours



Limonite and haematite debris



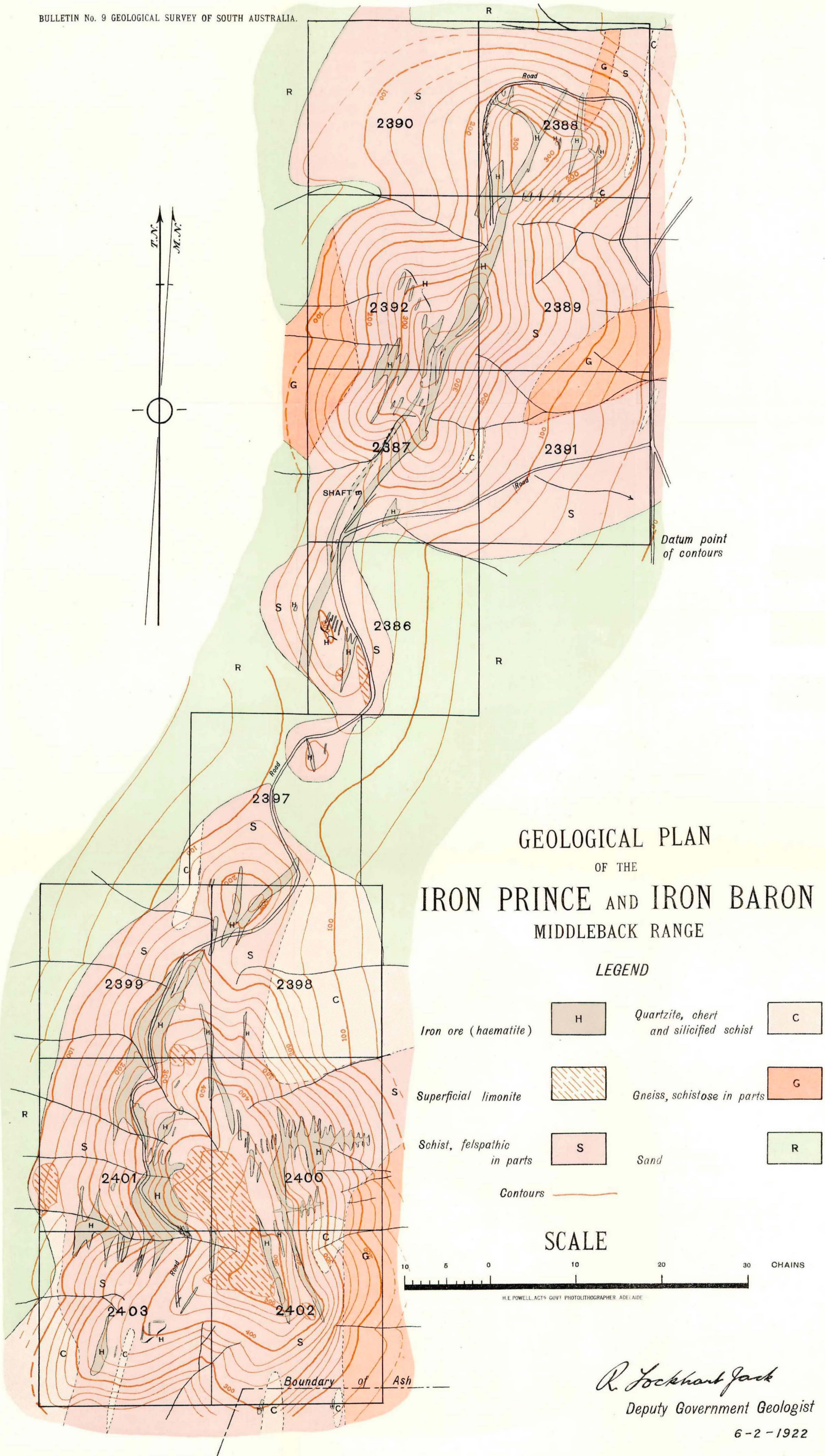
Superficial debris of the hill slopes
and alluvium of the flats



Tracks (other than roads)



R. Lockhart Jack
Deputy Government Geologist
6-2-1922



GEOLOGICAL PLAN
OF THE
IRON PRINCE AND IRON BARON
MIDDLEBACK RANGE

LEGEND

Iron ore (haematite)	H	Quartzite, chert and silicified schist	C
Superficial limonite		Gneiss, schistose in parts	G
Schist, felspathic in parts	S	Sand	R
Contours			

SCALE



R. Lockhart Jack
Deputy Government Geologist
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